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Disability and Rehabilitation: Assistive Technology

A Review Evaluating the Design and Standardisation of Rollators and Smart-Walkers to Improve Future Products

Submission ID	242439859
Article Type	Review - double anonymous peer review
Keywords	Walker, Rollator, Assistive Technology, i-HAAT, Standards, Design, Smart Walker
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A Review Evaluating the Design and Standardisation of Rollators and Smart-Walkers to Improve Future Products

Running Header: Rollator and Smart-Walker Evaluation

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Word Count (excluding abstract and references) = 10107;

Abstract word count = 237;

Number of tables in manuscript = 4;

Number of figures in manuscript = 3;

Appendix = 3

Number of references = 117

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Abstract

Four-wheeled walkers are widely used mobility aids for older adults. With the emergence of commercial smart-walkers, flaws in the design of rollators may carry over to the design of these emerging smart-walkers, necessitating a review of their design and standardisation. This review aims to evaluate how health, design, regulatory, research, and user factors intersect in the development and implementation of rollators and smart-walkers. It seeks to identify gaps in current standards, literature, and commercial products, with the goal of informing more user context-sensitive design and implementation approaches.

The findings highlight limitations in current standardisation, regulatory and risk management frameworks for emerging commercial smart-walkers for personal and home use, and significant scope for improving rollator design to better reflect diverse user contexts, including device functionality, usability, and education. Additionally, the findings show that rollators and smart-walkers must evolve to adopt an inclusive design approach that prioritises ergonomic considerations, usability, safety, and the user's contextual relevance.

This review underscores the importance of collaboration among health professionals, regulators, researchers, designers, and users to enhance rollator efficacy, usability, and safety. Future efforts in rollator and smart-walker design should incorporate structured feedback mechanisms and standardised evaluation tools to assess how users interact with devices and documentation. By addressing these factors, this review offers insights to guide the future design and implementation of rollators and smart-walkers, ensuring they better meet the needs, preferences and contexts of diverse users in personal and home settings.

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Introduction

There is a growing need for assistive technologies that support independent living and physical activity in an aging population (Carlsson & Lundälv, 2022; Mansouri & Goher, 2016). Walkers, particularly four-wheeled rollators, are widely used by older adults with gait or balance impairments to improve mobility and reduce fall risk (Arefin *et al.*, 2020; Lindemann *et al.*, 2016; Mundt *et al.*, 2019; O'Brien *et al.*, 2021; Schmucker *et al.*, 2024; Silva *et al.*, 2019; Van Hook *et al.*, 2003). However, studies have highlighted fall risks associated with using walkers in some settings (Carlsson & Lundälv, 2022; Costamagnal *et al.*, 2019; Johnson *et al.*, 2022; Lindemann *et al.*, 2016; Mansouri & Goher, 2016; van Riel *et al.*, 2014), with severe injuries associated with older adults using rollators (Carlsson & Lundälv, 2022; Lindemann *et al.*, 2016; van Riel *et al.*, 2014). These risks often stem from incorrect device usage or loss of balance during use (Carlsson & Lundälv, 2022; Costamagnal *et al.*, 2019; Johnson *et al.*, 2022; Lindemann *et al.*, 2016; Mansouri & Goher, 2016; van Riel *et al.*, 2014) and represent a growing public health concern exacerbated by an aging population (Carlsson & Lundälv, 2022; Grimmer *et al.*, 2019; Johnson *et al.*, 2022; Liu *et al.*, 2009; Mansouri & Goher, 2016; Silva *et al.*, 2019). Despite the diversity in commercially available rollators, there is limited research into their usability and user satisfaction (Schmucker *et al.*, 2024). Additionally, few studies have systematically examined the complex interactions between assistive technology, its usage, and the user context, leaving gaps in the holistic understanding of assistive technology (Lenker, 2021).

Walkers generally consist of a rigid frame with four legs, with variants defined by the International Organisation for Standardisation (ISO) as walking frames (with no wheels), front-wheel walkers (ISO, 2021a) and four-wheeled rollators (ISO, 2021b, 2024). Rollators, in particular, are widely used as home or personal assistive technology. Walker variants exist for different health needs (Jayaraman *et al.*, 2019) and, more recently, rollators with enhanced electronics and smart features (Frizera-Neto *et al.*, 2011), commonly known as smart-walkers (Chang *et al.*, 2021; Dune *et al.*, 2012; Moreira *et al.*, 2019; Mostofa *et al.*, 2021; Orenius *et al.*, 2021; Wernera *et al.*, 2018; Zhao *et al.*, 2020), are emerging for home or personal use.

Standards and regulations govern the design of rollators, defining device quality, safety, and accessibility (Hobson *et al.*, 2006). Gaps in these frameworks may contribute to the lack of

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consensus on device efficacy, safety and implementation (O'Brien *et al.*, 2021; van Riel *et al.*, 2014), and inherent gaps in rollator design may carry through to emerging commercial smart-walker development, potentially exacerbating health risks associated with using rollators. Unclear definitions of accessibility within standards and policies add further challenges, creating inconsistencies between their goals and actual outcomes in assistive product design (Persson *et al.*, 2015).

This review is positioned within a broader *Interdependent Human Activity Assistive Technology* (i-HAAT) (Lee *et al.*, 2024; Lee *et al.*, 2020) framework to consider the intersecting facets of rollator and smart-walker design and user contexts. Given the dearth of systematic tools for assessing the usability of rollators (Schmucker *et al.*, 2024), the i-HAAT theoretical framework provides a structured approach to analysing the complex interactions among four interdependent domains: the human (individual user), activities performed by the user, the assistive technology (rollator or smart-walker), and the biopsychosocial context of device usage in which these domains interact (Lee *et al.*, 2024; Lee *et al.*, 2020). This review contributes to the broader body of i-HAAT research by concentrating on the assistive technology domain, advancing a holistic understanding of commercial rollator and smart-walkers systems.

While substantial research exists on experimental and clinical smart-walkers, this review focuses specifically on commercially available smart-walkers intended for home and personal use, alongside commercial rollators. It evaluates their designs from a person-centred perspective, emphasising alignment with user contexts. The review systematically assesses the current landscape of standards, regulations, and research literature relating to rollators and smart-walkers. The aim is to explore how these elements intersect and whether they align with usability research and real-world user needs. By identifying inconsistencies in classifications, design criteria, and documentation practices, and examining how product features correspond to varied user contexts, this review highlights gaps in current approaches and offers insights to inform the future development of more effective, context-sensitive rollators and smart-walkers for personal and home use.

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Method

This review adopts the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) guidelines (Page *et al.*, 2021) and the Moriaty (2011) Benchmarking framework to systematically evaluate the conformity (adherence to standards) and compliance (adherence to broader regulations) frameworks that govern the classification, design and testing of rollators (Hobson *et al.*, 2006). Using the International Organisation for Standardisation (ISO) 11199-2 rollator standard as the *exemplar*, the review analyses the *causal relata*, including relevant standards, regulations and existing commercial products, to determine alignments, discrepancies and their influence on the design of products, and identify gaps for improvements to future commercial rollators and smart-walkers (Moriarty, 2011).

An initial systematic search of standards and the broader literature (Figure 1) identified **(i) terminologies** (Schmucker *et al.*, 2024) and classifications (Hobson *et al.*, 2006) for various walkers (including rollator) and smart-walkers (Table 1). These findings guided the subsequent review of relevant standards, regulatory frameworks, and commercially available rollators and smart-walkers.

The *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) (Page *et al.*, 2021) guided the systematic search (Figure 1) of: **(ii) standards** (Table 2) from Australian Standards (AS, 2015), the American Society for Testing and Materials (ASTM, 2017), British Standards (BS, 1987), the Bureau of Indian Standards (BIS, 1991), the European Norms Standards (DIN-EN, 1999), the International Electrotechnical Commission (IEC, 2022), the International Organisation for Standardisation (ISO, 2022), and Japanese Industrial Standard (JIS, 2012); **(iii) regulations** (Table 3) for walker standards conformity and regulatory databases under the Australian Therapeutic Goods Administration (TGA, 2021) and the Australian Register of Therapeutic Goods (ARTG, 2024), the European Commission (EC, 2021) Database on Medical Devices (EUDAMED, 2024), and the United States (US) Federal Drugs Administration (FDA, 2024a); **(iv) product specifications and documentation** for the commercial rollators and smart-walkers included in this review (Figure 3, Table 4; Appendix A, B1 and B2 contain the full datasets). The product searches were conducted in 2023, and the standards and regulatory database searches were updated in December 2024.

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Moriarty's Framework – Data Collection and Analysis

Moriarty's (2011) framework consists of six levels **(BT0 to BT5)**; this review encompasses *(**BT0**, Moriarty's first stage)* examining the standards that directly govern the classification, design and testing (Hobson *et al.*, 2006) of rollators and smart-walkers using ISO 11199-2 as the exemplar (Moriarty, 2011); *(**BT1**, Moriarty's second stage)* understanding the causal relata and identifying alignments and discrepancies through the analysis of relevant standards, regulations, product features and specifications, and product documentation; and *(**BT2**, Moriarty's third stage)* identifying improvements through analysis and synthesis of the exemplar **(BT0)** and the causal relata **(BT1)**, focusing on commercial rollators and smart-walkers for personal and home use.

In stages **(BT0)** and **(BT1)**, *primary standards* (Table 2) that directly govern the classification, design, and testing of all walkers are analysed, followed by a review of *secondary standards* (Table 2) related to the design of all walkers and smart-walkers. The analysis utilised three databases: the American Society for Testing and Materials (ASTM) Compass (accessed 22/12/2024), Engineering Workbench by Accuris (accessed 22/12/2024), and Techsheet Enterprise (accessed 22/12/2024). Additionally, the review examined current regulations and regulatory databases (Table 3) to identify regulated commercial walkers and evaluate standards conformity and regulatory compliance requirements under the Australian Therapeutic Goods Administration (TGA), the European Database for Medical Devices (EUDAMED), and the United States Food and Drug Administration (FDA).

As part of **(BT1),** a search was conducted to identify commercial rollators and smart-walkers for personal and home use, listed on mobility aid manufacturer and supplier web pages. Product features, specifications, and manufacturers' documentation were analysed to determine alignments and discrepancies, conforming or relating to standards.

Finally, in stage **(BT2)**, which focuses on identifying improvements to future design, the review analyses the exemplar and causal relata, including primary and secondary standards (Table 2, Figure 2), regulatory frameworks (Table 3), research literature, and commercial rollator and smart-walker specifications (Figure 3) and documentation (Table 4). The findings are synthesised in the discussion section, identifying key areas for improvement to future product design of rollators and smart-walkers.

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Inclusion and Exclusion

Standards

a) Walker design and testing standards (DIN-EN, ISO) reviewed encompass walking frames and front-wheeled walkers (ISO, 2021a), rollators (DIN-EN, 1999; ISO, 2021b, 2024), table or platform walkers (ISO, 2005), and walking trolleys (ISO, 2019b). The analysis of the ISO 11199-2, rollators standard, includes its updated 2024 amendment (ISO, 2024).

b) Ergonomic standards (ISO, 2021c) encompassing anthropometric and ergonomic data affecting device usability and safety were included. Normative references within current walker standards were excluded.

c) Walker variants, *three-wheeled, knee,* and *reverse walkers* referenced in ISO standards (ISO, 2021b), were beyond the scope of this review and therefore excluded.

d) The examination of Technical Specifications (IEC, 2022) is included to identify areas for further standardisation of smart-walkers, acknowledging the current absence of separate established standards for these devices.

Review of Regulations

e) This review provides an overview of walker regulations under the Australia Therapeutic Goods Administration (TGA, 2021), the United States Federal Drugs Administration (FDA, 2024c), and the European Commission Database on Medical Devices (EUDAMED, 2023). An in-depth analysis of the detailed product traceability processes is beyond the scope of this review.

Review of Commercial Rollators and Smart-walkers

f) This review examines rollators and smart-walkers that are commercially available for personal and home use, excluding those designed for clinical settings or research applications.

g) The review focuses on rollators (ISO, 2021b, 2024), rollators resembling walking trolleys (ISO, 2019b), and commercial smart-walkers as assistive technology for home and personal use, available in the public domain, using the terms identified (Table 1).

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h) Review of products comprises features, specifications and documentation identified in (g, above). The analysis of physical products is outside the scope due to product accessibility.

i) Review of instruction information ISO 11199-2: 2021, section 20.3 comprises manufacturers' user guides, specifications and spare parts lists for download in the public domain on the manufacturer and supplier websites for the products in (g, above). The analysis of information, maintenance, and services available upon request are outside the scope.

Results

This review presents four main results aligned with Moriarty's (2011) framework (**BT0-BT2**):

i. **(BT0)** Identification of walker and smart-walker terminologies (Table 1) to support the search and selection of relevant standards (Table 2).

ii. **(BT1)** Analysis of standards relevant to walker and smart-walker design and testing (Figure 2).

iii. **(BT1)** Evaluation of the causal relata through a review of compliance frameworks and databases across Australia (AU), Europe (EU), and the United States (US) (Table 3).

iv. **(BT1)** Evaluation of the causal relata through a review of commercial rollators and smart-walkers, including features, specifications (Figure 3; Appendix A contains the full data), and manufacturers' documentation (Table 4; Appendices B1 and B2 contains the full data).

These findings inform **(BT2);** a synthesis of the exemplar and causal relata discussing the follow results sections.

Results i.(a): Terminology and Definitions

Thirty one varying terms were identified across standards and literature, including terms for walkers (n=20), smart-walkers (n=5), and broader descriptors such as assistive technology or walking aids (n=6). These terms aligns with a four-level classification hierarchy: assistive technology (ISO, 2022), walking aids, walkers, and smart-walkers (Table 1).

Within the walker category, device types delineats into walking frames, front-wheeled walkers (ISO, 2021a), rollators (ISO, 2021b), forearm walkers (ISO, 2005), and walking trolleys

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(ISO, 2019b). Smart-walkers include both sensor-enhanced navigation walkers (IEC, 2022) and powered rollators (ISO, 2022). These classifications follow ISO 9999:2022, a recognised international standard that aligns with the World Health Organisation's International Classification of Functioning, Disability and Health (WHO, ICF) (ICF, 2022). A summary of these varying terms and classification are in (Table 1)

Assistive technology intended for non-healthcare indoor or outdoor use in home and personal settings is the broader classification encompassing walkers (ICF, 2022; ISO, 2022). Walkers are considered low-risk, Class I Medical Devices under the Australian (AU), European Commission (EU), and United States (US) regulatory frameworks (ARTG, 2024; EUDAMED, 2023; FDA, 2024c), and should conform to the requirements stipulated in ISO Standards (Figure 2).

Walkers

Standard walkers (Frizera-Neto *et al.*, 2011; Sehgal *et al.*, 2021; Van Hook *et al.*, 2003) are also known as *walking frames with no wheels* (ISO, 2021a), *zimmer frames,* and *fixed walkers* (Silva *et al.*, 2019), has four legs with rubber tips and requires lifting during use. They provide the best stability and balance while walking or standing (Sehgal *et al.*, 2021), supporting full body weight (ISO, 2021a). However, they require a controlled, slower gait pattern and greater physical effort (Silva *et al.*, 2019) and are suitable for moderate to severe lower extremity impairments (Van Hook *et al.*, 2003).

Front-wheeled walkers (Frizera-Neto *et al.*, 2011; ISO, 2021a; Van Hook *et al.*, 2003) have front wheels and rubber tipped rear legs. They are intermediaries between standard walkers and rollators (four-wheeled walkers). They support full-body weight (ISO, 2021a) and suit patients with difficulty lifting (Silva *et al.*, 2019; Van Hook *et al.*, 2003), allowing for partial lifting and pushing.

Rollators (Frizera-Neto *et al.*, 2011; Sehgal *et al.*, 2021; Van Hook *et al.*, 2003), also called *mobility walkers* (BIS, 1991), *rolling walkers, four-wheeled walkers* (Hunter *et al.*, 2020)*,* and *seat walkers* (Prihartadi *et al.*, 2023), are designed to support users with moderate mobility limitations who benefit from a more dynamic gait. Unlike standard or front-wheeled walkers, rollators facilitate faster, more natural movement (Dune *et al.*, 2012; Hansson *et al.*, 2022; Silva *et al.*, 2019) and are

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better suited to higher-functioning persons with decreased endurance and minimal weight-bearing needs (Sehgal *et al.*, 2021), prioritising ease of use over maximum stability (Frizera-Neto *et al.*, 2011; Silva *et al.*, 2019). Beyond ambulation support, rollators can assist with daily activities (Brännström *et al.*, 2012; Mansouri & Goher, 2016; Sehgal *et al.*, 2021). Typical features include a foldable frame, handgrips or crossbar, four wheels, a seat, storage attachments, and optional accessories such as cane or cup holders (ISO, 2021b). The specifications and features of the 98 commerical rollators included in this review are summarised in Figure 3; included in Appendix A is a full list of models classified under general use (n=72), indoor use (n=9), walking trolley (n=2), off-road use (n=9), dual wheelchair or transit use (n=6), Parkinson's disease smart-walker (n-2), and powered rollator (n=6).

Forearm walkers* and *walking tables (ISO, 2005) are a variation of rollators and walking frames. These walkers promotes upright posture and greater gait efficiency with better upper body forearm support, and are better suited to ambulation (Atoyebi *et al.*, 2019; Jayaraman *et al.*, 2019). Forearm walker standards (ISO, 2005) are included in this review, but commercially available forearm walker themselves are beyond the scope of this review.

Walking trolleys (Asano *et al.*, 2019) share similarities with rollators, featuring handles for pushing and pulling, provides support for walking, standing and sitting, and a storage bag for daily activities. However, they are designed for users who can walk independently and differ from rollators in that users stand further behind the device. They also have additional testing requirements to accommodate higher *safety working loads*. The assessment of walking trolley standards (ISO, 2019b) and two commercially available walking trolleys are included in this review.

Dual use *wheelchair* or *transit rollators* with dual-use functionality, generally conform to standards for rollators (ISO, 2021b, 2024) and sometimes wheelchairs (ISO, 2014a). Limited literature exists on these walker variants. Dual use rollators are Included in the product review, but wheelchairs themselves are beyond the scope of this review.

Smart Walkers

Smart-walkers, also known as *robotic or intelligent walkers*, are rollator-based mobility aids enhanced with electronics such as sensors, actuators, and navigation systems. They enable

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different levels of support, including fully autonomous, semi-autonomous, or manual assistance to improve gait, navigation, and user safety (IEC, 2022; Moreira *et al.*, 2019). While smart-walkers have been widely researched over the past decades (Frizera-Neto *et al.*, 2011; Lindemann *et al.*, 2016; Saveriano *et al.*, 2021; Sequeira, 2019), this review focuses on two commercially available categories for home and personal use, including *powered rollators* (ISO, 2022) and smart-walkers aiding *Parkinson's disease* (Moustris *et al.*, 2021), as well as the navigation and planning concept outlined in IEC Technical Specifications 63134: Active Assisted Living (AAL) Use Cases (IEC, 2022) (Figure 2).

Powered rollators with guided support (Chang *et al.*, 2021; Frizera-Neto *et al.*, 2011; Moreira *et al.*, 2019; Wernera *et al.*, 2018; Zhao *et al.*, 2020) incorporate robotic enhancements to assist with uphill and downhill mobility, braking, lateral stability, and in some cases, gait and strength training (Moustris *et al.*, 2021; Sato *et al.*, 2019; Wernera *et al.*, 2018). This review includes six *powered rollators* (ISO, 2022) with guided support functionality (Figure 3).

Parkinson's disease *smart-waker* (Moustris *et al.*, 2021) integrate sensors, lasers, and sonars to facilitate gait coordination and reduce freezing episodes. They provide cues for movement, improve motor control, and support balance (Dune *et al.*, 2012; Frizera-Neto *et al.*, 2011; Moreira *et al.*, 2019)*.* Two commercial models with Parkinson's-specific features were identified in this review (Figure 3).

Navigation and planning *smart-walkers* feature Global Positioning Systems (GPS), obstacle detection, and haptic/audio feedback to support users with visual or mobility impairments in journey planning and navigation (IEC, 2022; Mansouri & Goher, 2016; Moreira *et al.*, 2019; Moustris *et al.*, 2021; Orenius *et al.*, 2021; Palopoli *et al.*, 2015; Wernera *et al.*, 2018; Zhao *et al.*, 2020). Some are designed to encourage social and physical activity (Jimenez *et al.*, 2022). No commercial devices with these capabilities were identified, but a definition is articulated within IEC TS 63134.

Health monitoring smart-walkers are well-documented in research, integrating biometric sensors or connect to wearables to monitor vital signs such as heart rate, blood pressure, and energy expenditure (Lindemann *et al.*, 2016; Moreira *et al.*, 2019; Orenius *et al.*, 2021; Silva *et al.*,

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2019). However, no commercially available models or standardisations were identified during this review.

Results i.(b): Standards for Walkers, Rollators and Smart-walkers

This review identified and extracted: (a) primary standards (n=12) for the design and classification of walkers; and (b) secondary standards (n=17) referencing quality, risk management, ergonomics, and a smart-walker Technical Specification. The final analysis comprises primary standards (n=6) (DIN-EN, 1999; ISO, 2005, 2019b, 2021a, 2021b, 2022, 2024) and secondary standards (n=5) (IEC, 2022; ISO, 2009, 2017, 2018, 2019a, 2021c) after excluding superseded (n=6), normative references (n=8) and standards not relevant to walker design and testing (n=4) (Table 2, Figure 2).

Results ii: Analysis of Walker Standards

This review examined five walker-related standards, using ISO 11199-2: 2021 (Rollators) as the exemplar, which includes an amendment to, alongside four causal relata: ISO 11199-1: 2021 (Walking Frames), ISO 11199-3: 2005 (Walking Table), ISO 19894: 2019 (Walking Trolleys), and the European Norm DIN-EN 1985. All ISO standards follow a consistent format, with ISO 11199-1 and ISO 11199-2 offering the most comprehensive guidance (Figure 2). DIN-EN 1985 supplements ISO requirements by specifying additional European conformity requirements (DIN-EN, 1999) (Figure 2). Key requirements relevant to function, usability and safety are summarised in (Figure 2). Additionally, risk and quality management standards are elaborated in the subsections below.

Risk and Quality Management

ISO walker standards reference ISO 14971 *Application of Risk Management for Medical Devices* (ISO, 2019a) and ISO 12100 *Risk Management of Mechanical Safety and Hazards* (ISO, 2010) for mechanical safety and hazard control, including for devices intended to operate in conjunction with others (ISO, 2019b, 2021a, 2021b). ISO 14971 outlines processes for risk analysis, evaluation, mitigation, and monitoring across the product lifecycle. Annex A of ISO

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11199-1 and ISO 11199-2 lists key risk management considerations for walking frames and rollators (Figure 2).

ISO 13485 Medical Devices Quality Management Systems, Requirements for Regulatory Processes, offers a global standard for regulatory compliance and continuous improvement across the product lifecycle, from design to post-market surveillance (ISO, 2017). Although ISO 13485 supports the application of ISO 14971 in areas such as design, production, and post-production planning (ISO, 2019a), ISO walker standards do not explicitly require conformity with ISO 13485.

ISO/TR 22411: Ergonomics Data

ISO/TR 22411: 2021 provides standardised ergonomics data used to inform other standards (ISO, 2014b, 2021c). It includes data related to the use of rollators: sensory characteristics (vision, hearing, touch, thermal sense), physical characteristics (strength, dexterity, mobility, body size), grip diameter, upper body movement (fine motor skills, reach, wrist rotation), lower body movement (walking speed). Some available data sets highlight the effects of age, gender and the impact of anthropometric variances of different world regions. Notably, the walker standards analysed in this review (DIN-EN, 1999; ISO, 2005, 2019b, 2021a, 2021b, 2022, 2024) do not capture the anthropometric data from ISO/TR 2241 (Figure 2).

IECTS 63134 Ed. 1.1 Active Assisted Living (AAL) Use Cases

Although the term ‘powered rollator’ is defined in ISO 9999 there is currently no ISO standard specifying design requirements for powered rollators. The International Electrotechnical Commission (IEC), which develops global standards for electrical and electronic technologies, provides a Technical Specification for a navigational smart-walker under IEC TS 63134 – AAL Use Cases (IEC, 2022). AAL is a fluid term for *Ambient-Assisted-Living* and *Active-Assistive-Living* encompassing Information Technology (IT) enabled healthcare products to support independent living for older adults and persons with disabilities (Aumayr, 2016). Section B.5 of this document provides a foundation for further standardisation, including an overview of a navigational smart-walker equipped with sensors and internet connectivity to assist individuals in safely returning home. It outlines the system design, including functionality, risk profile, IT topography, and data security (Figure 3).

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Results iii: Regulations of Rollators Across Different Regions

This review compares regulatory frameworks in Australia (AU), the European Commission (EU), and the United States (US) to highlight differences in conformity, compliance, terminology, and traceability for rollators. There were (n=25, AU), (n=743, EU), and (n=100, US) registered walkers identified under each of these (TGA, EUDAMED, FDA) databases in 2024. While similar low risk classifications (Class I) were identified, descriptive terminology varied across the regulator bodies (TGA, Basic walker; EUDAMED, Wheeled walker; FDA, Mechanical walker) (Table 3).

The Therapeutic Goods Administration of Australia (TGA)

The TGA is the Australian regulatory body that administers the *Therapeutic Goods Act 1989*, including the (Excluded Goods) Determination (TGA, 2021), which defines rollators as assistive technology under Schedule 1: 2B. Rollators registered on the Australian Register of Therapeutic Goods (ARTG, 2024) (n=25) are described as foldable mechanical walkers and are classified as Class I Medical Devices (Table 3). While not requiring a formal conformity assessment certificate, these devices must still meet essential safety and performance principles, implied through risk management processes (TGA, 2021).

The European Databank on Medical Devices (EUDAMED)

Rollators in Europe must comply with European Medical Devices Regulation (EU MDR 2017/745) and obtain CE marking as *Class I non-invasive assistive devices* (EC, 2021). Unique Device Identification (UDI) is required for traceability (EC, 2020). Prior to the launch of the European Databank on Medical Devices (EUDAMED, 2024), device data were distributed across multiple registers. For example, in Germany, Medical Technical Aids Register (HMV, 2023) listed (n=114) rollators, while the Federal Institute for Drugs and Medical Devices (BfArM, 2023) tracked device recalls. EUDAMED currently lists (n=743) entries under category Y120606: *Wheeled walker* (Table 3).

The United States (US) Food and Drugs Administration (FDA)

The FDA defines rollators as *Class I Devices, mechanical walkers for physical medicine* under CFR Title 21, section 890.3825 (FDA, 2024a). Walkers are exempt from *510(k) premarket*

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notifications. Standards conformity assessment is implied (FDA, 2024c) via quality management obligations, including record keeping (CFR Title 21, section 820.180) and complaint handling (CFR Title 21, section 820.198). A search of the FDA Establishment Registration & Device Listing (FDA, 2024b) identified (n=100) registered entries under *Walker, mechanical* (Table 3).

Results iv: Commercial Rollators and Smart-walkers

Searching commercially available products using terminologies from (Table 1) identified commercial products (n=106) on mobility aid manufacturers' and suppliers' web pages (n=30). The features and specifications for rollators (n=98), powered rollators (n=6), and smart-walkers for Parkinson's disease (n=2) were analysed (Figure 3; Appendix A). From these, some products provided documents for download (n=83), comprising instruction manuals and spare parts lists. The final review included the manufacturer's documentation for rollators (n=70), which included retrofitted smart-walkers for Parkinson's disease (n=2) (Table 4; Appendices B1 and B2).

Product Review

A search of (n=30) manufacturer and supplier websites identified (n=106) commercial products, including rollators (n=98), powered rollators (n=6), and retrofitted smart-walkers for Parkinson's disease (n=2). Of these, (n=70) provided downloadable documentation used in the final product review (Table 4; Appendices B1–B2).

Using ISO 11199-2 as the exemplar (Figure 2), product specifications and features were analysed across conformity criterias such as use environment, adjustability, and braking. Rollators were primarily designed for general (n=72), indoor (n=9), or outdoor/offroad (n=10) use. Additional variants included walking trolleys (n=2), dual-use wheelchair or transit walkers (n=6), and smart-walkers (n=8), comprising powered rollators (n=6) and retrofitted Parkinson's disease smart-walkers (n=2) (Figure 3; Appendix A).

The product review revealed considerable variation in rollator features and partial alignment with ISO 11199-2 standards. Key findings (Figure 3) include:

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a) **Brake types** were predominantly lever (n=60) and pad (n=28), with limited use of drum or disc systems; electric and resistance brakes were excluded due to inconsistent documentation.

b) **Handgrip and brake ergonomics** showed variability in grip span (62 mm to 77 mm), with (n=22) models featuring tilted ergonomic handles.

c) **Handle height adjustability** vary between 630 mm and 1120 mm, mostly conforming to ISO limits, though one model slightly exceeded the 25 mm increment threshold (byAcre, 2023).

d) **Indoor rollators** (n = 9) showed key differences from general and outdoor models, including narrower dimensions, smaller wheels, some exceeded DIN-EN 1985's minimum 600 mm width. Notably, indoor rollators provided a waist height tray, and (n=8 of 9) lacked seating.

e) **Seat dimensions** and **wheel diameters** varied widely, with larger wheels typically found in offroad models.

f) **Storage loading** ranged from 2 kg to15 kg (general rollators) to 25 kg (walking trolleys).

g) **Weight** exceeded 10 kg for heavy-duty, offroad, and smart-walker models, raising lifting and transportability concerns under ISO 11199-2 guidelines.

h) **Wheel materials** included polyurethane, thermoplastic rubber, and pneumatic options, affecting ride comfort and terrain suitability.

Product Documentation Review

From the (n=106) products reviewed, (n=83) provided downloadable documentation, of which (n=70) non-duplicate manuals were included for analysis of standards conformity (Table 4; Appendices B1–B2). Exclusions included universal duplicates (n = 11), non-adjustable walkers (n = 1), and powered rollators lacking required detail (n = 2). The analysis focused on alignment with ISO 11199-2, Section 20.3 (instruction manuals).

Most manuals complied with basic conformity requirements such as maximum user mass and storage loading, maintenance and folding instructions, adjustment mechanisms and control

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locations, and warranty and contact details. However, non-conformities were identified (Table 4) in areas concerning safe and informed use:

a) Use Environment and Dimensions: Many manuals failed to specify intended environments (e.g. indoor/outdoor use), full product dimensions, or safe working load of accessories.
b) Assembly and Disassembly: Instructions were often incomplete or missing.
c) Warnings and Safety Advice: Several manuals lacked warnings for device-specific risks (e.g. hot surfaces, uneven ground, or seating precautions).
d) Cleaning and Disinfection: Limited or inconsistent hygiene instructions were noted across documents.

Specific non-conformities by clause inclue:

e) 20.3(a), accessible formats: None of the manuals provided alternate formats for users with visual, cognitive, or reading impairments. Manufacturer contact for alternate versions was outside the review scope.
f) 20.3(j), component weight >10 kg: No manuals (n = 0) listed weights for dismantable parts exceeding 10 kg, potentially relevant for dual-use or smart-walker models.
g) 20.3(k), use with other devices: Only (n=2) smart-walker manuals addressed compatibility or safety considerations when retro fitted rollators are used alongside other (electronic) devices.
h) 20.3(n), serial number location: Traceability documentation varied with only (n=37) manuals specifying where to find the device ID or serial number, and some manufacturers published this online or included it on the product itself (byAcre, 2023; Rollz Mobility, 2024).

Beyond ISO conformity, the depth and clarity of instructions varied considerably:

i) Most documents (n = 66) were in booklet formats, of which (n=58) provided graphical information. A small number (n = 4) offered only minimal leaflet-style content with limited instructional value, lacking in written or illustrated guidance for device usage, safety and maintenance.

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j) Maintenance guidance coverage ranged from basic cleaning (n = 57) to advice on servicing (n = 47) and structured maintenance plans (n = 7).

k) Only a subset of manuals addressed ergonomic and usage safety, (n = 26) showing hand height positioning, (n = 5) showing elbow angle guidance, (n = 28) showing seating safety, (n = 16) showing inclines or slope use cautions, some with specific lateral stability or tipping limits (side angle: n = 5; multi-directional: n = 1)

Discussion

This discussion synthesises the key findings from the analysis of exemplar devices (BT0) and causal relata (BT1), aligned with Stage (BT2) of Moriarty's (2011) benchmarking framework, to identify opportunities for improving rollator and smart-walker design and their governing frameworks. While rollators play an essential role in supporting mobility and independence, current products often fall short of addressing the diverse contexts, preferences, and capabilities of users. The findings reveal inconsistencies between international standards, commercial implementations, and user needs. To advance safety, usability, and adoption, this discussion is structured into four interrelated domains: (i) limits of current standards, (ii) real-world design gaps, (iii) regulatory misalignment, and (iv) inclusive design implications. Each section offers targeted insights to support the development of more effective and user-responsive mobility devices.

i. Limits of Current Standards

This section examines how existing ISO and associated standards address critical ergonomic, performance, and documentation elements of walker design. Although standards like ISO 11199-2 establish fundamental characteristics, significant deficiencies persist in aspects such as handle ergonomics, brake mechanisms, weight limits, seating arrangements, inclination safety, and instructional materials. Some specifications may depend on obsolete or limited anthropometric data, reducing global applicability. The findings indicate the inadequacies of standards in addressing varied user profiles and real-world conditions, and the need for for justified and risk managed non-conformity where appropriate.

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Handlebar Height Ergonomics

The handlebar height of the current products under review is generally consistent with the anthropometric parameters outlined in ISO TR 22411. Research suggests that the ideal handlebar height should range from 48% to 55% of an individual's height (Edelstein, 2019; Takanokura, 2009), with higher ratios suited for older adults with poor balance (Choi et al., 2015; Takanokura, 2009). The ergonomic handlebar range of about 707 mm to 1077 mm is derived by applying this ratio to the ISO TR 22411 height data for the 5th to 95th percentile, which ranges from 1473 mm to 1959 mm (ISO, 2021c) (Figure 2). This range is accommodated by the commercial products that were reviewed (Figure 3; Appendix A), which cater to a wide range of user heights.

However, there are discrepancies. ISO 11199-2 specifies a height range for hand control of 800 mm to 1200 mm, which corresponds with the 5th to 95th percentile elbow-to-hand anthropometric limits (889 mm to 1239 mm) (Figure 2), but not with the optimal handlebar height derived from overall body height. ISO 11199-1 specifies a minimum handlebar height of 900 mm, beyond the minimum optimal height of 707 mm, identified above. Furthermore, a Scandinavian rollator (byAcre, 2023) had a height adjustment increase of 30 mm, beyond the ISO 11199-2 standard limit of 25 mm (Figure 2), potentially attributable to Dutch users in the 95th percentile. Handlebar height affects user posture and musculoskeletal health (Guo et al., 2020; Liu et al., 2009), and these discrepancies suggest that strict conformity to ISO requirements may not be universally appropriate.

Handle Ergonomics

The ISO 11199-2 handgrip diameter range of 20 mm to 50 mm (Figure 2) aligns with ISO TR 22411 anthropometric data derived from Dutch populations, which report optimal diameters of 36–40 mm for women and 39–43 mm for men (ISO, 2021c). However, this range diverges from findings in broader literature. While a Taiwanese study reported an optimal grip diameter of 47.6 mm (Liao, 2016), US-based studies report the most comfortable grip spans as 50–55 mm for small to medium hands, 55–65 mm for larger hands, and narrower spans for older adults (Kong *et al.*, 2007; Lee *et al.*, 2009; Yan & Downing, 2013). These studies suggest that the standardised grip

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specifications may not align with regional or individual comfort preferences, an aspect of rollator design that could benefit from adjustability to better suit individual users.

Torque sensitivity is significantly affected by grip span, particularly for individuals with musculoskeletal impairments, despite having minimal impact on net grip strength (Liao, 2016; McDowell et al., 2012). Additionally, only 22 of the 98 products reviewed (Figure 3; Appendix A) had articulated handgrip angles that supported the recommended elbow flexion angle of 15˚ to 30˚ (Bradley & Hernandez, 2011; Van Hook et al., 2003). These ergonomic features are particularly relevant for users with wrist limitations.

ISO 11199-2 does not specify ergonomic parameters for handle grip, articulation, or brake grip span, limiting its relevance to real-world rollator use. Factors such as grip shape, size, angle, material, and surface texture significantly affect comfort, usability, and function (Halim *et al.*, 2020; Kong *et al.*, 2007; Lee *et al.*, 2009; Liao, 2016; Walsh & Petrie, 2016). The variability in research findings and product designs highlights a misalignment between standardised guidelines and the needs of users, particularly those with wrist impairments or lower functional capacities, reinforcing evidence that anthropometric data alone are insufficient to define optimal grip ergonomics for diverse populations (Sternäng *et al.*, 2015), warranting further consideration in future product design.

Braking Efficacy

Future rollator design can benefit from improvements to the efficacy and usability the braking system. ISO 11199-2 sets a maximum brake grip span of 75 mm and a maximum pulling force of 40 N, which are both aligned with US bicycle brake standards and the minimum force capacity of a 5th percentile female hand (Code of Federal Regulations, 2021; Marras & Karwowski, 2021). However, while current ISO standards specify brake span and operation force limits, there is no requirement for manufacturers' documentation to report product-specific grip spans or operational force, with only two European manufacturers disclosing brake spans of 62–77 mm (byAcre, 2023; Topro, 2024).

Ineffective braking is a key failure mode linked to falls (Finkel *et al.*, 1997). Studies into rollators and smart-walkers suggest that braking capability is influenced by user strength,

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coordination, and reaction time, which are particularly relevant to ageing and vulnerable populations (Frizera-Neto *et al.*, 2011; Hirotomi, 2021; Lee *et al.*, 2009; McDowell *et al.*, 2012). Most products reviewed used lever (n=60) or pad (n=28) brakes, with some using drum (n=8) or disc (n=1) brakes, which improves braking in wet conditions (Rollz Mobility, 2024). Recent smart-walker studies suggest that improving braking efficacy reduces user anxiety and fear of falling, allowing them to focus on walking (Hirotomi, 2021). These findings highlight the need for further research into improving braking systems that considers user capabilities (Hirotomi, 2021) and usage environments, such as wet or uneven surfaces (Carlsson & Lundälv, 2022; Finkel *et al.*, 1997; Hansson *et al.*, 2022; Spijkerman, 2022), while also considering the impact of overall device weight and any associated modifications.

Product Weight Exceeding 10 kg

Some commercial rollators and smart-walkers, such as heavy-duty, outdoor, dual-use transit rollators, powered models, and retrofitted smart-walkers, surpass the ISO walker standard's 10 kg (one person) lift restriction due to their reinforced frames and additional components such as batteries and electronics (Frizera-Neto et al., 2011). The majority of products reviewed weigh less than 10 kg; lightweight designs are preferred for their portability, particularly when lifting the device in and out of cars, which is a significant accessibility concern (Brandt et al., 2003; Finkel et al., 1997).

The risk of injury in older individuals is increased by overestimating their lifting capacity. For example, whereas 75% of elderly Canadians believed they could lift 50 kg, half encountered difficulties with 10 kg in everyday activities (Kopec et al., 2017). Risk is further influenced by lift distance, as injury risk increases when a rollator's centre of gravity is positioned further from the body (Safe Work Australia, 1990). This risk factor varies by gender and impacts users' ability to load devices into vehicles (Brandt et al., 2003; Brännström et al., 2012). Heavier rollators also impede curb and slope navigation (Costamagnal et al., 2019; Lindemann et al., 2016). Therefore, added functionality must be reconciled with user safety and practical usability, ensuring that rollators and smart-walkers remain appropriate for their intended daily use (Ang et al., 2019).

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Seating Height and Position

Seat height affects the rollator's stability, usability, and comfort. However, it varies widely across products, standards, and research literature. ISO 19894 specifies a minimum seat height of 350 mm, which is consistent with the ISO/TR 22411 5th percentile sitting height of 333 mm. While lower seats may be more comfortable, research suggests that higher seat heights, ranging from 538 mm to 630 mm, facilitate easier sit-to-stand transitions (Edelstein, 2019; Finkel et al., 1997). The review identified (n=32) distinct seat heights across products, ranging from 460 mm to 640 mm. The height of the seat influences user comfort as well as the stability and functionality of the rollator, necessitating careful placement within the frame design (Finkel et al., 1997). Additionally, under-seat storage may impede leg movement and obstruct safe usage (Spijkerman, 2022) and natural gait patterns (Edelstein, 2019). These findings highlight the need for further research into how rollator seats, including height, positioning, and storage design, can affect comfort, usability, and stability.

Slope and Tilt Angles

ISO standards specify higher minimum tilt stability angles for outdoor walkers, emphasising stability across different environments (ISO, 2005, 2021a, 2021b). However, there are no explicit requirements to include incline safety information in instruction manuals. Fewer than half of the instruction manuals reviewed provided incline warnings, and only six provided specific incline limits (limit on all sides, n=1; side incline limits, n=5). Recent studies suggest that loading of rollators, such as groceries in the storage bag, can affect stability depending on the task and user impairments, particularly when walking backwards, navigating obstacles, and stepping up or down (Costamagnal *et al.*, 2019). These variations underscore a need to improve the standardisation of information for device incline safety and warrant further research on rollator loading and stability in varied daily use contexts beyond ideal usage conditions (Costamagnal *et al.*, 2019; Dune *et al.*, 2012).

Wheel Sizes, Materials and Usage Environments

The analysis of wheel materials (Figure 3; Appendix A) reveals a balance between durability and grip. Harder wheels (n=10, PU, PUR, PVC) offer longer wear life but reduced

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traction, while softer materials (n=22, TPE, TPR, EVA, silicone) improve comfort and grip across varied surfaces. Pneumatic rubber wheels (n=6) provide superior shock absorption but wear more quickly. Some walkers (n=4) used a combination of front (PU) and rear wheel materials (EVA, TPR) to balance performance across different terrains. These distinctions highlight a need for tailoring material selection to use environments, balancing comfort (Finkel et al., 1997) and durability (Amin & Amin, 2011). However, current ISO walker standards do not account for how wheel materials perform across real-world conditions. Testing is confined to plywood surfaces and does not account for wet, uneven, or slippery settings (ISO, 2021b), presenting a gap in design validation for outdoor and mixed-terrain applications.

Wheel diameter varied among commercial products. While ISO standards recommend minimum front-wheel diameters of >75 mm for indoor and >180 mm for outdoor use, some products (n=22) combined larger front wheels (200 mm to 285 mm) with smaller rear wheels (180 mm to 235 mm). Of the products reviewed that provided wheel diameters (n=104), over half used 200 mm wheels for both front and rear. Larger wheels improve obstacle navigation and terrain adaptability (Rentschier *et al.*, 2003), which supports their use in off-road and outdoor models. These variations suggest that wheel size meaningfully impacts stability, safety, and usability (Spijkerman, 2022).

ISO 11199-2 includes general risk management guidance on surface conditions and intended use environments (ISO, 2021b), but does not offer detailed specifications for wheel profile, tread patterns, or drainage, which are factors addressed in ISO 24415-1 for rubber tips and DIN-EN 1985 for walking aids (DIN-EN, 1999; ISO, 2009). Standards and testing protocols need to be expanded to encompass surface-dependent performance of wheels for rollators and smart-walkers, alongside more explicit manufacturer information for users and clinicians on their appropriate use and maintenance requirements throughout the device's lifespan (Carlsson & Lundälv, 2022; Finkel et al., 1997; Hansson et al., 2022; Spijkerman, 2022).

Instruction Manual Improvements

Despite the benefits of rollators, their use is linked to injuries and fall risks, especially among frail users (Costamagnal et al., 2019; van Riel et al., 2014). These risks have been partially

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attributed to inadequate education and improper usage, underscoring the need for improved instructions, training, and maintenance to mitigate risks (Carlsson & Lundälv, 2022; Costamagnal et al., 2019; O'Connor et al., 2017; Oldfreya et al., 2024; Schmucker et al., 2024; van Riel et al., 2014; Walsh & Petrie, 2016).

The review of (n=70) rollator instruction manuals identified inconsistent conformity to ISO documentation requirements. Although most manuals included general usage, safety, and maintenance, the uniformity and depth of content varied significantly, few (n=11) manuals included maintenance plans, and many lacked detailed usage guidance essential for user safety. Key information such as optimal wrist positioning, posture alignment (Costamagnal et al., 2019; Guo et al., 2020; Liu et al., 2009), elbow flexion angles (Bradley & Hernandez, 2011; Van Hook et al., 2003) and tilt stability limits (ISO, 2005, 2021a, 2021b) are typically not included; information that would facilitate proper device utilisation and initial selection.

While ISO standards require inclusion of safety, usage, and maintenance content (ISO, 2005, 2021a, 2021b), they do not specify the quality, clarity, or comprehensiveness of the information. As a result, instruction manuals may achieve basic conformity while offering limited practical value. Considering the essential role of user education in reducing injury risk (Carlsson & Lundälv, 2022; Costamagnal et al., 2019; O'Connor et al., 2017; Oldfreya et al., 2024; Schmucker et al., 2024; van Riel et al., 2014; Walsh & Petrie, 2016), improving the clarity, consistency, and usability of instruction manuals should be prioritised to ensure safe and effective device use.

ii. Real World Gaps

Despite meeting conformity, many walker and rollator products diverge from ideal usability when evaluated in real-world contexts. This section examines product features and design considerations insufficiently addressed by existing standards, including aesthetic factors, walking variants (walking trolley), the indoor–outdoor distinction, and a lack of standardisation for emerging commercial smart-walkers. These gaps highlight the need to embed context-sensitive, functional, and psychosocial considerations into future design and standardisation efforts.

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Aesthetics in Device Design

The inclusion of aesthetics considerations in the European Norms walker standard DIN-EN 1985, absent in ISO standards, impacts product acceptance and user experience, which can influence motivation and self-esteem (Brännström *et al.*, 2012; Orenius *et al.*, 2021; Palopoli *et al.*, 2015; Renda *et al.*, 2016; Spijkerman, 2022; Suhaimi *et al.*, 2022; Walsh & Petrie, 2016). Balancing functionality with subjective user preferences is challenging for product design. For example, while forearm walkers provide better support and posture (Atoyebi *et al.*, 2019; Jayaraman *et al.*, 2019), their design may be unappealing to some users, potentially leading to stigmatisation and reduced usage (Atoyebi *et al.*, 2019; Brännström *et al.*, 2012; Walsh & Petrie, 2016). Balancing functionality, usability, and safety with individual requirements and the nuanced role of preferences in rollator design requires further investigation. Incorporating user preferences into the design process is crucial to ensure these devices are effective, accepted, and embraced by users, a principle supported by the World Health Organisation, United Nations International Children's Emergency Fund Global Report on Assistive Technology (WHO-UNICEF, 2022), which acknowledges that assistive technology influences users' sense of identity, agency, and self-presentation (Lee *et al.*, 2024; Walsh & Petrie, 2016).

Walking Trolley and Bag Loading Stability

There are several critical differences between ISO 11199-2 (rollators) and ISO 19894 (walking trolley). ISO 19894 requires higher minimum tilt angles, longer static load tests, and additional handle torque and seat stability tests. Walking trolleys are intended for users who can walk independently (ISO, 2019b), and its clinical evaluation recommendations, beyond ISO 11199-1, -2, and -3 risk management guidelines, underscore gaps in standardisation for further consideration. ISO 19894 requires manufacturers to declare maximum user mass and Safety Working Load (SWL), incorporating both into stability and strength testing (ISO, 2019b). In contrast, ISO 11199-2 assumes SWL is relative to user weight and does not require its separate nomination (ISO, 2021b). This difference highlights distinctions in user weight and SWL limits and their relevance to rollators with larger bag capacities.

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This review identified two rollators resembling walking trolleys represented in ISO 19894 but conforming to ISO 11199-2 in the product documentation. These rollators have a higher accessory bag capacity (12 kg to 25 kg) than outdoor rollators (4 kg to 12 kg), which predominantly provide a bag capacity of 5 kg (n=25). A study into the effects of loading distribution on stability found that a loading distribution and an increase of 5.5 % or more of the user's weight slows down gait by a third, increasing fall risk (Shin et al., 2015). Added weight can present safety risks that may impact the device's braking (Finkel et al., 1997) and stability (Costamagnal et al., 2019). To better understand the risks and long-term health implications of using rollators with increased SWL and bag-loading capacities in daily use, as well as the influence of additional load on braking performance on inclines (Asano et al., 2019), additional research is required. These insights could also be used to inform the development of new testing requirements for future revisions to ISO standards.

Indoor and Outdoor Rollators

The design and function of indoor and outdoor rollators vary significantly, as indicated in ISO and EN walker standards (DIN-EN, 1999; ISO, 2005, 2019b, 2021a, 2021b, 2024) (Figure 2). Key differences include (a) a minimum wheel diameter of 180 mm for outdoor designs; (b) a smaller minimum diameter of 75 mm for indoor use; (c) higher tilting stability requirements for outdoor use; (d) a maximum width of 600 mm for indoor use as stipulated in DIN-EN 1985, consistent with studies recommending smaller rollators for space-constrained environments (Atoyebi et al., 2019; Brännström et al., 2012); and (e) varying minimum outdoor wheel depth specifications between ISO (22 mm) and DIN-EN 1985 (28 mm). Additionally, the majority of indoor rollators examined in this review (n=8 of 9) do not feature seats (Figure 3; Appendix A), indicating design goals that differ from those of outdoor models.

While some studies supported lightweight designs for indoor use (Spijkerman, 2022), one manufacturer (byAcre, 2023) promotes a heavier-weight indoor design (Scandinavian Butler) for increased stability, consistent with studies suggesting higher fall risk indoors (Johnson et al., 2022; Mansouri & Goher, 2016), especially during morning and evening transitions between standing and sitting (Mansouri & Goher, 2016), highlighting the need to balance stability with manoeuvrability in

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device design. Indoor and outdoor rollators are highly contextually dependent in their functionality and usability (Brännström et al., 2012; Carlsson & Lundälv, 2022; Prihartadi et al., 2023; Walsh & Petrie, 2016). However, the distinctive functions, risks, and usability of these two categories have not been comprehensively investigated (Schmucker et al., 2024). Future research should examine the impact of these disparities on the safety and use of indoor and outdoor rollators, including their practical applications and specific requirements. This includes the integration of user perspectives (O'Brien et al., 2021), which are often overlooked assistive technology design (Lee et al., 2024).

Standardisation of Smart-walkers

This review identified two definitions for smart-walkers and one for retrofitted rollators: the design for a navigation smart-walker in IEC TS 63134 AAL Use Cases (IEC, 2022), the ISO classification for *powered rollators* (ISO, 2022), and rollators with smart features working as a system (with other devices) (ISO, 2021b), such as the Rollz Motion, Rhythm (Rollz Mobility, 2024), and Topro Neuro (Topro, 2024). These definitions provide limited coverage for the other smart-walkers functionalities identified in this review: guided (robotic) support (Chang *et al.*, 2021; Frizera-Neto *et al.*, 2011; Moreira *et al.*, 2019; Wernera *et al.*, 2018; Zhao *et al.*, 2020), gait monitoring (Dune *et al.*, 2012; Frizera-Neto *et al.*, 2011; Moreira *et al.*, 2019), Parkinson's disease (Moustris *et al.*, 2021), health monitoring (Lindemann *et al.*, 2016; Moreira *et al.*, 2019; Orenius *et al.*, 2021), and navigation and planning (IEC, 2022; Mansouri & Goher, 2016; Moreira *et al.*, 2019; Moustris *et al.*, 2021; Orenius *et al.*, 2021; Palopoli *et al.*, 2015; Wernera *et al.*, 2018; Zhao *et al.*, 2020). These limitations in current definitions highlight the need for expanded standardisation to encompass other smart-walker functionalities.

The shift from Ambient Assisted Living to Active Assisted Living within the IEC TS 63134 AAL technology ecosystem emphasises a focus on individual user needs (Eklund et al., 2019; Jarl & Lundqvist, 2018; Lee et al., 2020; WHO-UNICEF, 2022) and user empowerment through technological assistive technology solutions (Aumayr, 2016). However, IEC TS 63134 (IEC, 2022) only provides a foundational information technology framework for one type of smart-walker, lacking conformity criteria for specifications, ergonomics, testing, or risk management. While ISO provides a classification for powered rollators, this review identified no related standards.

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The Major Criticality rating (IEC, 2022) assigned to the navigation smart-walker in IEC TS 63134 emphasises the significant risks associated with its failure or misuse. Despite this, commercial smart-walker' risk profiles remain ambiguous. For example, the powered rollator presented by Bao (2022) provided details of the walker's robotic system design (Bao *et al.*, 2022) but no details of clinical evaluation. Furthermore, there is a dearth of understanding of smart-walker efficacy (Sato *et al.*, 2019), particularly from the user's perspective (Wernera *et al.*, 2018). Given the emergence of commercial smart-walkers, this ambiguity and the existing lack of consensus on rollator efficacy (Schmucker *et al.*, 2024), safety, and implementation (O'Brien *et al.*, 2021; Prihartadi *et al.*, 2023; van Riel *et al.*, 2014) necessitate further evaluation of smart-walker classifications and standardisations.

iii. Regulatory Misalignment

This section analyses the alignment of regulatory systems in Australia, Europe, and the United States with international standards and user safety goals. While European and US systems mandate traceability and conformity through independent oversight, Australia's ARTG process allows optional registration and lacks clear ISO enforcement. These discrepancies impact risk and quality management across the product lifecycle. Leveraging ISO 13485 and ISO 14971 frameworks, alongside user training and maintenance infrastructure, is essential for harmonising global regulation and improving device reliability, particularly as new smart technologies emerge.

Regulations in Different Regions and Their Alignment with Standards

Regulatory frameworks across different regions have explicit or implied standards conformity requirements. The key differences in regulations and the relative databases were variances in traceability processes, underscored by a significant variance in registered walkers across the ARTG, EUDAMED, and FDA databases (Table 3). In Australia, the registration of assistive technology, including rollators, under the ARTG provided a self-regulated process with optional ARTG registration and implied standards conformity requirements. The European and US regulations emphasise more robust product traceability and safety management, with the EU mandate for the Conformité Européene (CE) mark verified through independent testing and the US mandate for manufacturer registration, documentation, and customer complaint records.

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The observed variances in regulatory frameworks and traceability processes across regions underscore current conformity barriers with global standards such as ISO 13485 for quality management (ISO, 2017) and ISO 14971 for risk management (ISO, 2019a). These disparities reflect Lenker's (2021) findings, calling for standardised, data-driven approaches to ensure assistive technology safety, effectiveness, and usability throughout the product lifecycle. These findings align with the WHO-UNICEF Global Report on Assistive Technology (WHO-UNICEF, 2022), which advocated for the closing of global regulatory and policy gaps, as well as improving accessibility to device maintenance, quality information (Carlsson & Lundälv, 2022), and training resources (Walsh & Petrie, 2016) for users, thereby improving the safety and effectiveness of assistive devices.

Risk and Quality Management in Standards

Manufacturers are responsible for managing risks under current ISO walker standards. However, ISO 14971 risk management standard (ISO, 2019a) does not establish acceptable risk levels; instead, it provides a framework for assessing risks through data review, literature, comparable products, and cross-disciplinary collaboration. Basic risk management and safety considerations are outlined in Annex A of the ISO walker standards (ISO, 2021a, 2021b). The ergonomic impacts related to ageing detailed in ISO/TR 22411 (ISO, 2021c), extend further into sensory and physical characteristics and anthropometric differences across age, gender, impairments, and global regions. Research is needed to determine how these differences can inform designs tailored to individual user needs in future rollator development (Mansouri & Goher, 2016; O'Brien et al., 2021; Prihartadi et al., 2023), and to understand the long-term effects of powered rollators providing guided support on users' gait (Sato et al., 2019). To improve future rollator designs and enable widespread commercial adoption of smart-walkers, adopting a holistic risk management framework integrating ISO 14971 with ISO/TR 22411 would ensure comprehensive ergonomic and safety considerations are addressed.

ISO 13485 for quality management (ISO, 2018) provides a framework extending beyond rollator standards to support broader regulatory compliance. Post-production monitoring is an integral part of risk management, addressing the need for effective maintenance and support

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infrastructure to mitigate risks associated with rollator use (Carlsson & Lundälv, 2022; Oldfreya et al., 2024). Incorporating user input into device development, evaluation, and monitoring, as emphasised by the WHO-UNICEF Global Report on Assistive Technology (WHO-UNICEF, 2022), can enhance responsiveness to individual users' current and future needs (Brandt *et al.*, 2003; Brännström *et al.*, 2012), ultimately improving products and systems.

Although current standards provide structural guidance, they remain limited in scope, focusing on generic user models and basic safety requirements without accounting for the diverse contexts of real-world use. Drawing on the Inter-dependent Human Activity Assistive Technology (i-HAAT) framework and the WHO-UNICEF Global Report on Assistive Technology, this synthesis argues for reorienting device development, evaluation, and support towards context-sensitive approaches that integrate user perspectives to enhance device usability and safety for diverse populations.

iv. Final Synthesis: Towards Inclusive and Person-Centred Design

This review identified four key domains: limitations in standardisation, real-world gaps, regulatory misalignments, and inclusive design. Together, these highlight disconnects between current frameworks and the lack of research on the usability of commercial rollators (Schmucker *et al.*, 2024) and smart-walker for home and personal use (Bao *et al.*, 2022; IEC, 2022; Sato *et al.*, 2019). These domains were examined through evidence-based analysis of:

a) Standardisation for walkers (DIN-EN, 1999; ISO, 2005, 2019b, 2021a, 2021b, 2024), ergonomics (ISO, 2021c), smart walker technical specification (IEC, 2022), risk and quality management (ISO, 2018, 2019a).
b) Ergonomic design parameters, including handle height (Edelstein, 2019; Takanokura, 2009), handle grip (Kong *et al.*, 2007; Lee *et al.*, 2009; Yan & Downing, 2013), braking efficacy (Hirotomi, 2021; McDowell *et al.*, 2012), product weight (Ang *et al.*, 2019; Kopec *et al.*, 2017), seat height (Edelstein, 2019; Finkel *et al.*, 1997), tilt angles (Costamagnal *et al.*, 2019), wheel design (Rentschier *et al.*, 2003; Spijkerman, 2022), and device instructions, including training and maintenance (Carlsson & Lundälv, 2022; Oldfreya *et al.*, 2024; Schmucker *et al.*, 2024);

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c) Real world gaps including aesthetics and stigma (Atoyebi *et al.*, 2019; Brännström *et al.*, 2012; Walsh & Petrie, 2016), rollator variants for indoor and outdoor use, walking trolleys (Asano *et al.*, 2019; Atoyebi *et al.*, 2019; Brännström *et al.*, 2012; Carlsson & Lundälv, 2022; Prihartadi *et al.*, 2023; Spijkerman, 2022; Walsh & Petrie, 2016);

d) Regulatory inconsistencies across Australia, EU, and US (Lenker, 2021; WHO-UNICEF, 2022), and the need for inclusive and context sensitive approaches (Lee *et al.*, 2024; Schmucker *et al.*, 2024; WHO-UNICEF, 2022) in design and system decision making.

While rollator standards provide a design framework, their alignment with broader definitions and goals of assistive technology may be limited (Persson et al., 2015). Standards often rely on generic assumptions, falling short in accounting for real-world complexities, and vary in regulatory oversight across regions. Inclusive design remains aspirational, with limited evidence on how users interact with commercial rollators and smart-walkers in daily life. Alongside the need to understand the complex dynamics of usability, the lack of empirical usability data (Schmucker et al., 2024) and minimal literature on emerging commercial smart-walkers, reinforce the need for research grounded in person-centred, context-sensitive approaches. These limitations, along with safety concerns arising from the lack of consensus on device efficacy and implementation, underscore the need to understand user needs, preferences, (Brännström *et al.*, 2012; Mansouri & Goher, 2016) and usage contexts (Jarl & Lundqvist, 2018; Lee *et al.*, 2024; Lee *et al.*, 2020), consistent with recommendations from the WHO-UNICEF Global Report on Assistive Technology (WHO-UNICEF, 2022).

These findings also reflect broader challenges in assistive technology development, where the shift from patient-centric to inclusive, person-centred design has yet to be fully embedded in practice (Alston & DeKerchove, 2023; Boger *et al.*, 2017; Lee *et al.*, 2024; Lee *et al.*, 2020; Phillips *et al.*, 2023). The i-HAAT framework (Lee et al., 2024; Lee et al., 2020) provides a conceptual foundation for designing products that reflect users' biopsychosocial realities, including physical, psychological, social, and environmental dimensions (Eklund *et al.*, 2019; Phillips *et al.*, 2023; WHO-UNICEF, 2022). To embed an inclusive design approach in rollator development, collaboration between stakeholders (ISO, 2018), including health professionals, regulators,

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researchers, designers, and rollator users, to enhance device efficacy and empower user autonomy, well-being, and dignity is required (Brandt *et al.*, 2003; Lee *et al.*, 2024; Lee *et al.*, 2020; Schmucker *et al.*, 2024; WHO-UNICEF, 2022).

Limitations

While this review uses the i-HAAT framework, it focuses exclusively on the assistive technology domain, aiming not to prescribe specific design outcomes or policies but to understand the key constraints, standards, and regulations that shape commercial rollators and smart-walkers for home and personal use. Future research exploring the interaction of assistive technology with the broader human and activity-related dimensions of rollators and smart-walkers within various contexts is warranted. Although this review identifies key gaps and offers critical insights, its findings are contextually bound, and a cautious approach should be considered when generalising.

The scope is limited to products and documentation accessible online through English-language websites. It excludes products or documentation in other languages. This focus narrows the generalisability of the findings and excludes detailed supplier-provided services. The regulatory and database analysis relies on frameworks such as ARTG, EUDAMED, and FDA, with variability in data availability potentially affecting the comprehensiveness of the findings. As such, the review contributes to how these frameworks influence the status quo of commercial products.

Finally, the review of rollator information manuals provides an overview of current documentation to identify gaps that impact user safety and device maintenance. However, this review is limited, as current walker standards do not include standardised tools for measuring the quality of the document contents, something that warrants future investigation (Gök *et al.*, 2019).

Future Research

This review provides insights into the current landscape of rollators and emerging smart-walkers. It examines the standards, regulations, and commercially available products, by comparing the exemplar (rollator standard) with the causal relata through a teleological process using Moriarty's benchmarking framework. Future research should integrate empirical data and standardised assessment tools to address dynamic interactions between users and devices. i-HAAT offers a guiding framework to ensure inclusive design remains integral to the research and

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design process. Finally, commercial smart-walkers require further classification and standardisation to account for their varied functionalities and risk profiles. Leveraging principles such as IEC TS 63134 AAL Use Cases and Active Assistive Living can guide future standardisation and design efforts, ensuring that technological solutions are effective, usable, safe, and contextually relevant.

Conclusions

Rollator use is expected to rise with an ageing population, highlighting ongoing debates around the efficacy, safety, and implementation of these devices. This review synthesised a broad and complex body of information and identified key limitations in current commercially available rollators and emerging smart-walkers, including limitations in standardisation, inconsistent classifications, fragmented design criteria, and poor alignment with ergonomic and contextual user needs. The findings also revealed a lack of standardised tools for evaluating documentation, weak alignment between product features and real-world usage contexts, and limited feedback mechanisms among users, health professionals, researchers, and designers.

These insights are relevant for improving future designs and informing implementation approaches that better reflect diverse user contexts. Variations in physical ability, environments, and activities are mirrored by a wide range of commercial rollator features and specifications. Despite the diversity in commercial rollators, these variations do not consistently correspond to users' real-world settings, physical capacities, or activity needs. Understanding this complexity is a critical first step towards bridging these gaps. Future rollator and smart-walker development must shift toward context-sensitive design approaches that embed individual user needs, preferences, and settings within product development, regulatory guidance, and implementation throughout the product lifecycle. Frameworks such as i-HAAT, which emphasise human context, offer useful starting points for this reorientation. Collaboration among stakeholders, such as health professionals, regulators, researchers, designers, and users, is needed. Breaking down siloed design processes is essential for achieving greater coherence between design outcomes and user contexts. Additionally, enabling feedback loops can further reconcile the disparity between designs and their practical application.

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Acknowledgements

The author thanks the co-authors for their invaluable contributions and dedication to this manuscript. The authors gratefully acknowledge the support from MedTechVic through their Impact Scholarship and to Swinburne University of Technology, Melbourne, Australia.

Disclosure Statement

The authors have no competing interests to declare.

Funding

This work was supported by the Swinburne University of Technology, Australia, MedTechVic Impact Scholarship.

Ethics and Consent

Not Applicable

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Table 1: Key Terminologies Used in the Search and Review of Standards, Regulations, Databases, and Commercial Rollators and Smart-Walkers

Classification	Instance	Terminologies/sub-classifications/variants from standards	Instance	Terminologies/synonyms from peer-reviewed literature
Assistive Technology	(1)	**Assistive Technology/product (O)** (ASTM, 2017; ICF, 2022; ISO, 2022)	(3)	Ambulatory Device (O) (Van Hook *et al.*, 2003)
	(2)	**Walking Aid** (**O)** (ISO, 2005)	(4)	Assistive Device (O) (Arefin *et al.*, 2020; O'Brien *et al.*, 2021)
			(5)	Mobility Assistive Device (O) (Arefin *et al.*, 2020)
			(6)	Mobliity Aid (O) (Prihartadi *et al.*, 2023)
Walking Aid	(1)	**Walker (W)** (Arefin *et al.*, 2020; Dune *et al.*, 2012)		
Walker	(2)	**Walking Frame (W)** (ISO, 2021a)	(9)	Zimmer Frame (W) (Carlsson & Lundälv, 2022)
			(10)	Regular/Standard Walker (W) (Van Hook *et al.*, 2003)
			(11)	**Front-Wheeled Walker (W)** (Van Hook *et al.*, 2003)
			(12)	Two-Wheeled Walker (W) (Dune *et al.*, 2012)
	(3)	**Rollator (W)** (Dune *et al.*, 2012; ISO, 2021b; Schmucker *et al.*, 2024)	(13)	**Four-Wheeled Walker (W)** (Van Hook *et al.*, 2003)
			(14)	Mobility Walker (W) (BIS, 1991; Hunter *et al.*, 2020)
			(15)	Rolling Walker (W) (Mundt *et al.*, 2019)
			(16)	Seat Walker (W) (Prihartadi *et al.*, 2023)
			(17)	Wheelie Walker (W) (Dune *et al.*, 2012; Hunter *et al.*, 2020)
Rollator	(4)	**Indoor/Outdoor Walker (W)** (Prihartadi *et al.*, 2023)		
Walking Frame Rollator	(5)	**Walking Table (W)** (ISO, 2005)		
	(6)	Platform Walker (W) (Mansouri & Goher, 2016)		
	(7)	**Forearm Walker (W)** (Jayaraman *et al.*, 2019; Mori *et al.*, 2021)		
Rollator (walker) variant			(18)	Knee Walker (W) (ISO, 2021b)
			(19)	Knee Scooter (W) (Mansouri & Goher, 2016)
			(20)	Three-Wheeled Walker (W) (Dune *et al.*, 2012; ISO, 2021b)
Walker	(8)	**Walking Trolley** (W) (ISO, 2019b)		
Smart-Walker	(1)	Smart Wheeled Walker (SW) (IEC, 2022)	(3)	Robotic Walker (SW) (Dune *et al.*, 2012)
	(2)	**Powered Rollator (SW)** (ISO, 2022)	(4)	**Smart-Walker (SW)** (Chang *et al.*, 2021; Dune *et al.*, 2012; Mansouri & Goher, 2016; Palopoli *et al.*, 2015)
			(5)	**Intelligent Walker** (**SW)** (Dune *et al.*, 2012)

Note 1 General:. Items in the table are organised by classification in order of assistive technology preceding walking aid, preceding walker, preceding walkers including rollator and smart-walker. **Bolded terminologies** are the focus of this review;
Note 2. Acronyms: (W) = Walker terminologies, **(SW) =** Smart-walker terminologies, (**O)** = Other relevant terminologies;
Note 3. Instances: Total terminologies for Walkers (n=20), Smart-walkers (n=5), Other relevant terminologies (n=6)

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Table 2: Standards and Technical Specifications relevant to the design of walkers, walker variations and smart-walkers

Terminology	Instance	Primary standards: (Directly governs classification, design and testing of walkers)	Instance	Secondary standards: (Related to the design and risk management of walkers and smart-walkers)
Assistive Technology	(7) **(1)**	AS ISO 9999:2018 (S); **ISO 9999:2022 (AT)** Assistive products - Classification and terminology (ISO, 2022)	(6) **(1)** (7) (8) (9) **(2)**	EN ISO 21856: 2022 (NR) AT (medical) – Requirements and test methods **ISO/TR 22411: 2021 (E)** Ergonomics data for use in the application of ISO/IEC Guide 71:2014 (ISO, 2021c) ISO IEC Guide 71: 2014 (NR) Guide for addressing accessibility in standards (ISO, 2014b) AS EN 12182: 2015 (N) Assistive products for persons with disability - requirements and test methods (AS, 2015) ISO 31000:2018 Risk Management Guidelines (NR) (TGA, 2021) **ISO 13485: 2017 (P)** Quality management system (ISO, 2017)
Walking Aid Ambulatory Device Assistive Device Mobility Aid Walker	**(2)** (8)	**DIN-EN 1985: 1998 (W)** Requirements and test methods (DIN-EN, 1999). BS 5104: 1974 (S) Adjustable height walking frames (BS, 1987) EN 1985	(10) **(3)**	AS/NZS ISO 11334.4 (NR)Wheelchair seating systems in motor vehicles **ISO-24415-1: 2009 (W)** Tips for assistive products for walking - Requirements and test methods - friction of tips (ISO, 2009)
Walking Frame Zimmer Frame Regular Walker Standard Walker **Front-Wheeled Walker**	**(3)** (9) (10)	**ISO 11199-1: 2021 (W)** Walking frame (ISO, 2021a) JIS T 9264:2012 (S) Based on ISO 11199-1:1999 (S) (JIS, 2012a)	**(4)**	**ISO 14971: 2019 (P)** Medical devices - Application of risk management to medical devices (ISO, 2019a)
Rollator **Four-Wheeled Walker** Mobility Walker Rolling Walker Seat Walker Wheeled/Wheelie Walker **Indoor Walker** **Outdoor Walker**	(11) **(4)** (12)	BIS 13017 (S) Walker Rollator (BIS, 1991) **ISO 11199-2: 2021 (W)** Rollators (ISO, 2021b), including the ISO 11199-2: 2024 amendment (ISO, 2024) JIS T 9265: 2012 (S) Based on ISO 11199-2 and 3 (JIS, 2012b)	(11) (12) (13) (14) (15) (16) (17)	ISO 8191-2 (N) Furniture - Ignitability of upholstered furniture ISO 10993-1 (N) Biological evaluation of medical devices ISO 13732-1 (N) Ergonomics of the thermal environment ISO 15223-1 (N) Medical device - Symbols, labelling, and information to be supplied ISO 20417 (N) Medical devices - Information to be supplied by the manufacturer ISO 7000 (N) Graphical symbols for use on equipment EN 614-1+A1 (N) Safety of machinery - Ergonomic principles
Walking Table **Forearm Walker**	**(5)**	**ISO 11199-3: 2005 (W)** Walking Tables		
Walking Trolley	**(6)**	**ISO 19894: 2019 (W)** Walking trolleys		
Smart-walker Robotic Walker **Powered Rollator**		None were identified	**(5)**	**IEC TS 63134 (SW)** – Technical Specification (navigation smart-walker) Active Assisted Living (AAL) use cases (IEC, 2022)

Note 1. General: (x) Bolded Standards are included in the final review (see inclusions and exclusion). Titles are paraphrased;
Note 2. Acronyms: AT = Assistive Technology, **E** = Ergonomics, **N** = Normative References, **NR** = Not Relevant, **P** = Quality and risk management processes, **W** = Walkers, **S** = Superseded, **SW** = Smart-walker; **Acronyms for standards: AS** = Australian Standard, **ASTM** = American Society for Testing and Materials, **BIS** = Bureau of Indian Standards, **BS** = British Standard, **DIN-EN** = Deutsches Institut für European Norms Standards, **IEC** = International Electrotechnical Commission, **ISO** = International Organisation for Standardisation, **JIS** = Japanese Industrial Standards;
Note 3. Instances: Total standards identified (n=29): Primary standards (n=12), Secondary standards (n=17); **Standards in the review**: Primary standards (n=6) and Secondary standards (n=5); **Excluded standards:** Superceded (n=6), normative standards (n=8), not related to walker design for home or personal use (n=4)

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Table 3: Summary of product regulatory bodies, traceability and registration of rollators and walkers

Registry Name	Classification	Compliances	Traceability	Total (Number) of Entries
Australian Register of Therapeutic Goods (ARTG, 2024)	Basic walker, foldable (Class I)	ISO Standards (implied)	Optional	(n=25) (ARTG, 2024) (Access date: 01/09/2024)
The European Database on Medical Devices (EC, 2021; EUDAMED, 2023, 2024)	Y12120505: Wheeled Walker (Class I)	ISO Standards (CE Certificate)	Unique Device Identification (UDI), manufacturer's details, documentation	(n=743) (EUDAMED, 2024) (Access date: 23/12/2024)
The German Medical Technical Aids Register (HMV, 2023)	Rollator (Class I)	ISO Standards (CE Certificate)	Follows European regulatory compliance	(n=114) (HMV, 2023) (Access date: 23/12/2024)
The US Food and Drug Administration (FDA, 2024b, 2024c)	Walker, Mechanical (Class I)	ISO Standards (implied)	Manufacturer's details, customer complaints, documentation,	(n=100) (FDA, 2024b) (Access date: 23/12/2024)

Note 1. General: Bolded registries are the focus of this review;
Note 2. Acronyms: CE = Conformite Europeenne, **EU** = European Union, **US** = United States, **ISO** = International Standard Organisation

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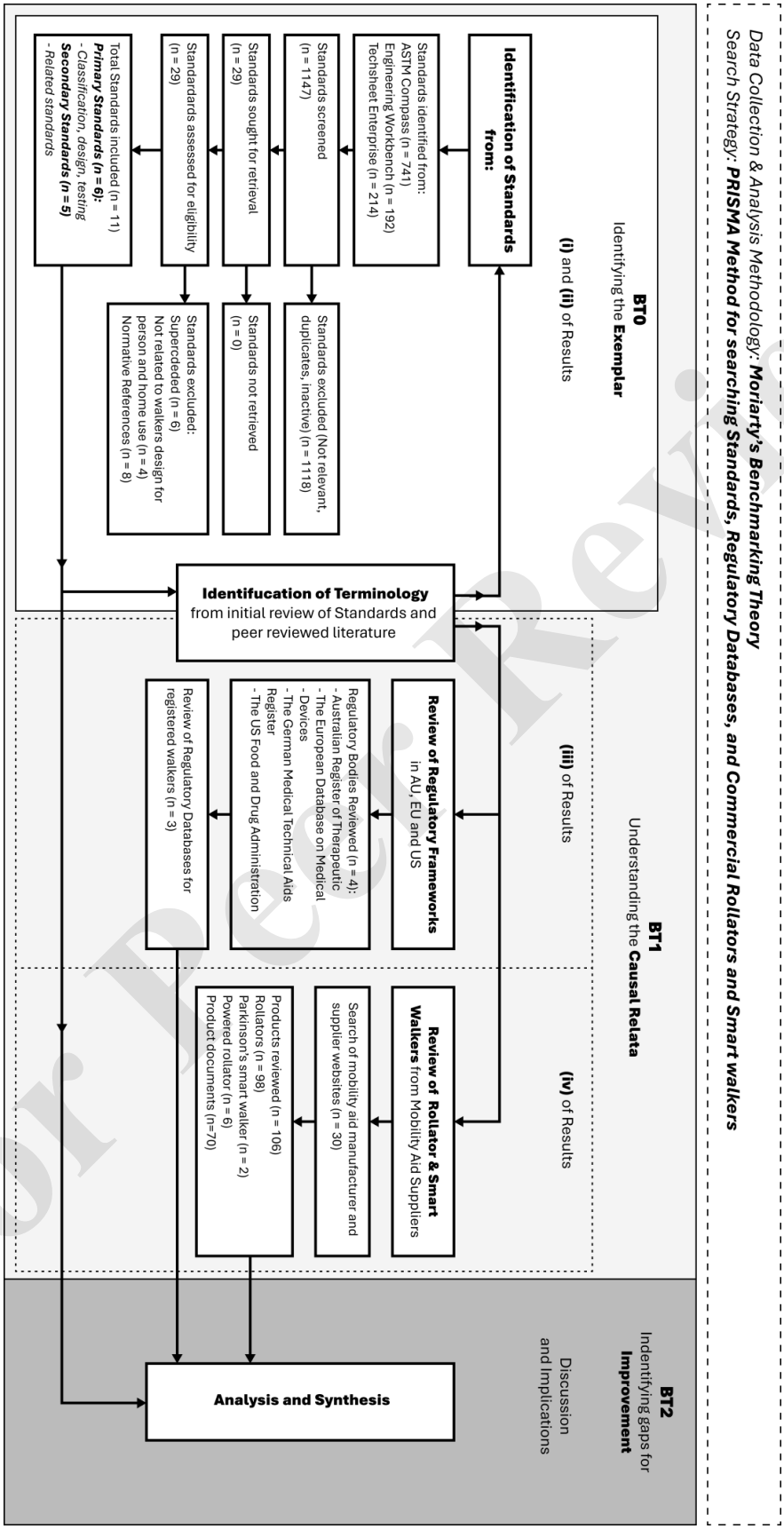
Table 4: Review of manufacturers' documentation: Summaries of conformity to Instructions Information requirements in standards

Requirements from ISO 11199 - 1: 2021 and ISO 11199 - 2: 2021, section 20.3 *Instruction manual*	**Instances**	**Notes**
a) Accessibility of information in formats for visual, reading or cognitive disabilities	(0)	Not found in documents
b) The planned usage, encompassing the target user and environment	(65)	Non-conformity (some)
c) Maximum user mass	(70)	Conformity (all)
d) The maximum safe weight limit for accessories used to carry loads (containers)	(49)	Non-conformity (some)
e) Minimum and maximum height of the rollator	(61)	Non-conformity (some)
f) Maintenance instructions, if applicable	(69)	Conformity (mostly)
g) Cleaning: If applicable, provide cleaning method, materials and anti-corrosion measures	(62)	Non-conformity (some)
h) Disinfection: If applicable, provide cleaning method, materials and corrosion details	(40)	Non-conformity (some)
i) Overall dimensions (width, length, and height) in mm, mass in kg, preparation for use; if applicable, folding or dismantling	(59)	Non-conformity (some)
j) Mass expressed in kg if the device is collapsable or contains removable parts over 10 kg	(0)	Relevant to some smart-walkers
k) Safe of device used in combination with other products and what they are, if applicable	(2)	Relevant to some smart-walkers
l) Manufacturer-approved accessories, detachable parts, and materials	(60)	Non-conformity (some)
m) Device folding or dismantling to assist in storage or transport;	(67)	Conformity (mostly)
n) The location and the type of identification for the unique product identification	(37)	Non-conformity (high)
o) Adjustments required before using the rollator and their impact on functionality	(69)	Conformity (mostly)
p) Details on adjustment options and required competency required to carry out the task	(69)	Conformity (mostly)
q) Operation of controls	(70)	Conformity (all)
r) Dismantling, re-assembly and information on any removable parts	(62)	Non-conformity (some)
s) Safe grip points for moving and handling components, dismantling, assembly or carrying	(69)	Conformity (mostly)
t) Surface temperature warning: changes due to external heat sources	(26)	Non-conformity (high)
u) Safety information: If the device usage or feature poses a hazard (e.g. holes, openings)	(62)	Non-conformity (some)
v) Safety information: If the device usage poses a hazard (e.g. squeezing)	(64)	Non-conformity (some)
w) How to obtain warranty information	(69)	Conformity (mostly)
x) Warning for fall risk (e.g. Incorrect use, do not use device to transport person)	(66)	Conformity (mostly)
Further analysis of document content	**Instances**	**Notes**
Documentation format: Booklets	(66)	Varying degrees of conformity
Documentation format: Leaflets	(4)	Minimal information, non-conformity
Documentation format: Graphical information	(58)	Graphical instructions
Maintenance information: Wear and tear	(57)	Parts wear and inspection advice
Maintenance information: Service advice	(47)	Self-service, supplier service advice
Maintenance information: Maintenance plan	(7)	Inspection and recommended parts
User safety: Hand height and positioning	(26)	Showing optimal hand position
User safety: Elbow flexion	(5)	Flexion angle relative to the handle
Slope or incline: General use	(16)	General use warnings
Slope or incline: When sitting	(28)	Safe use when sitting on the device
Slope or incline: Limit on all sides	(1)	All direction incline limit specified
Slope or incline: Sideways limit	(5)	Sideways incline limit specified

Note 1. General: This table presents a rephrased overview of requirements in section 20.3 Instruction manual. Refer to the specific standards for more detail;
Note 2. Instances: Total manuals (n=83); Manuals excluded (n=12): Universal duplicate manuals (n=11), non-adjustable rollator (n=1), powered rollator (n=2); Manuals reviewed (n=70): Rollator (n=70), including retrofitted Parkinson's (smart) walkers (n=2)
Note 3. Included in **Appendix B1 and B2** is a full list of manunuals reiviwed for rollator and smart walker models reviewed.

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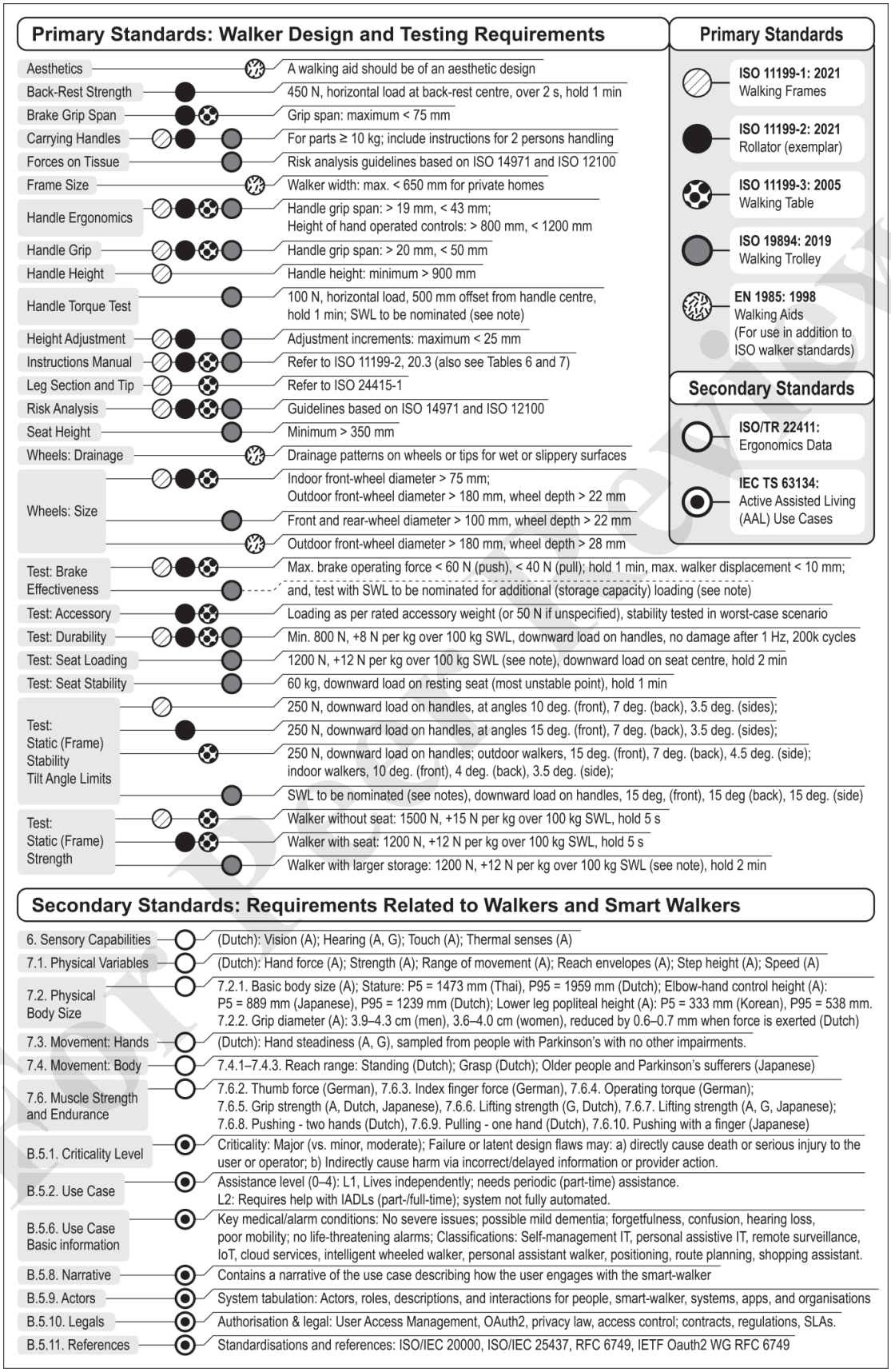
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Figure 1 Caption: Flow diagram of the search strategy, data collection and analysis methodology

Note: This review uses BT0, BT1, and BT2 levels of Moriarty's (2011) Benchmarking framework as the data collection and analysis methodology. The Preferred Reporting Items for Systematic Reviews and Meeta Analyses (PRISMA) (Page *et al.*, 2021) provide a systematic structure for the search method used to identify standards, regulatory databases, commercial rollators and smart-walkers.

Figure 1 Alt Text: A flow diagram outlining the systematic review process using PRISMA and Moriarty's benchmarking framework. It shows identification, screening, and analysis stages for standards (n=1147), regulatory databases, and commercial rollators and smart-walkers (n=106). The three benchmarking levels, BT0, BT1, and BT2, guide the analysis from exemplar identification to causal relationships and improvement gaps.

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Primary Standards: Walker Design and Testing Requirements
Aesthetics
A walking aid should be of an aesthetic design
Back-Rest Strength
450 N, horizontal load at back-rest centre, over 2 s, hold 1 min
Brake Grip Span
Grip span: maximum < 75 mm
Carrying Handles
For parts ≥ 10 kg; include instructions for 2 persons handling
Forces on Tissue
Risk analysis guidelines based on ISO 14971 and ISO 12100
Frame Size
Walker width: max. < 650 mm for private homes
Handle Ergonomics
Handle grip span: > 19 mm, < 43 mm;
Height of hand operated controls: > 800 mm, < 1200 mm
Handle Grip
Handle grip span: > 20 mm, < 50 mm
Handle Height
Handle height: minimum > 900 mm
Handle Torque Test
100 N, horizontal load, 500 mm offset from handle centre, hold 1 min; SWL to be nominated (see note)
Height Adjustment
Adjustment increments: maximum < 25 mm
Instructions Manual
Refer to ISO 11199-2, 20.3 (also see Tables 6 and 7)
Leg Section and Tip
Refer to ISO 24415-1
Risk Analysis
Guidelines based on ISO 14971 and ISO 12100
Seat Height
Minimum > 350 mm
Wheels: Drainage
Drainage patterns on wheels or tips for wet or slippery surfaces
Wheels: Size
Indoor front-wheel diameter > 75 mm;
Outdoor front-wheel diameter > 180 mm, wheel depth > 22 mm
Front and rear-wheel diameter > 100 mm, wheel depth > 22 mm
Outdoor front-wheel diameter > 180 mm, wheel depth > 28 mm
Test: Brake Effectiveness
Max. brake operating force < 60 N (push), < 40 N (pull); hold 1 min, max. walker displacement < 10 mm;
and, test with SWL to be nominated for additional (storage capacity) loading (see note)
Test: Accessory
Loading as per rated accessory weight (or 50 N if unspecified), stability tested in worst-case scenario
Test: Durability
Min. 800 N, +8 N per kg over 100 kg SWL, downward load on handles, no damage after 1 Hz, 200k cycles
Test: Seat Loading
1200 N, +12 N per kg over 100 kg SWL (see note), downward load on seat centre, hold 2 min
Test: Seat Stability
60 kg, downward load on resting seat (most unstable point), hold 1 min
Test: Static (Frame) Stability Tilt Angle Limits
250 N, downward load on handles, at angles 10 deg. (front), 7 deg. (back), 3.5 deg. (sides);
250 N, downward load on handles, at angles 15 deg. (front), 7 deg. (back), 3.5 deg. (sides);
250 N, downward load on handles; outdoor walkers, 15 deg. (front), 7 deg. (back), 4.5 deg. (side);
indoor walkers, 10 deg. (front), 4 deg. (back), 3.5 deg. (side);
SWL to be nominated (see notes), downward load on handles, 15 deg, (front), 15 deg (back), 15 deg. (side)
Test: Static (Frame) Strength
Walker without seat: 1500 N, +15 N per kg over 100 kg SWL, hold 5 s
Walker with seat: 1200 N, +12 N per kg over 100 kg SWL, hold 5 s
Walker with larger storage: 1200 N, +12 N per kg over 100 kg SWL (see note), hold 2 min
Primary Standards
ISO 11199-1: 2021
Walking Frames
ISO 11199-2: 2021
Rollator (exemplar)
ISO 11199-3: 2005
Walking Table
ISO 19894: 2019
Walking Trolley
EN 1985: 1998
Walking Aids
(For use in addition to ISO walker standards)
Secondary Standards
ISO/TR 22411:
Ergonomics Data
IEC TS 63134:
Active Assisted Living (AAL) Use Cases
Secondary Standards: Requirements Related to Walkers and Smart Walkers
6. Sensory Capabilities
(Dutch): Vision (A); Hearing (A, G); Touch (A); Thermal senses (A)
7.1. Physical Variables
(Dutch): Hand force (A); Strength (A); Range of movement (A); Reach envelopes (A); Step height (A); Speed (A)
7.2. Physical Body Size
7.2.1. Basic body size (A); Stature: P5 = 1473 mm (Thai), P95 = 1959 mm (Dutch); Elbow-hand control height (A): P5 = 889 mm (Japanese), P95 = 1239 mm (Dutch); Lower leg popliteal height (A): P5 = 333 mm (Korean), P95 = 538 mm.
7.2.2. Grip diameter (A): 3.9–4.3 cm (men), 3.6–4.0 cm (women), reduced by 0.6–0.7 mm when force is exerted (Dutch)
7.3. Movement: Hands
(Dutch): Hand steadiness (A, G), sampled from people with Parkinson's with no other impairments.
7.4. Movement: Body
7.4.1–7.4.3. Reach range: Standing (Dutch); Grasp (Dutch); Older people and Parkinson's sufferers (Japanese)
7.6. Muscle Strength and Endurance
7.6.2. Thumb force (German), 7.6.3. Index finger force (German), 7.6.4. Operating torque (German);
7.6.5. Grip strength (A, Dutch, Japanese), 7.6.6. Lifting strength (G, Dutch), 7.6.7. Lifting strength (A, G, Japanese);
7.6.8. Pushing - two hands (Dutch), 7.6.9. Pulling - one hand (Dutch), 7.6.10. Pushing with a finger (Japanese)
B.5.1. Criticality Level
Criticality: Major (vs. minor, moderate); Failure or latent design flaws may: a) directly cause death or serious injury to the user or operator; b) Indirectly cause harm via incorrect/delayed information or provider action.
B.5.2. Use Case
Assistance level (0–4): L1, Lives independently; needs periodic (part-time) assistance.
L2: Requires help with IADLs (part-/full-time); system not fully automated.
B.5.6. Use Case Basic information
Key medical/alarm conditions: No severe issues; possible mild dementia; forgetfulness, confusion, hearing loss, poor mobility; no life-threatening alarms; Classifications: Self-management IT, personal assistive IT, remote surveillance, IoT, cloud services, intelligent wheeled walker, personal assistant walker, positioning, route planning, shopping assistant.
B.5.8. Narrative
Contains a narrative of the use case describing how the user engages with the smart-walker
B.5.9. Actors
System tabulation: Actors, roles, descriptions, and interactions for people, smart-walker, systems, apps, and organisations
B.5.10. Legals
Authorisation & legal: User Access Management, OAuth2, privacy law, access control; contracts, regulations, SLAs.
B.5.11. References
Standardisations and references: ISO/IEC 20000, ISO/IEC 25437, RFC 6749, IETF Oauth2 WG RFC 6749

For Review

Figure 2 Caption: Review of standards: Summaries of key differences between walker standards and sections relevant to walkers and smart-walkers in other standards

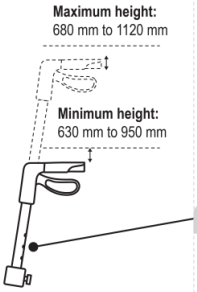
Note 1. General: This figure presents a redacted summary of standards relevant to rollator and smart-walker analysis; ISO 11199-2: 2021 includes the 2024 amendment (ISO, 2024); Static load tests typically use 100 kg user mass with ±2% tolerance; **SWL** (Safe Working Load) must be nominated and tested for walking trolleys with additional load-bearing specifications (ISO, 2019b).
Note 2. Acronyms: DIN-EN = Deutsches Institut für European Norms, **IEC** = International Electrotechnical Commission, **ISO** = International Organisation for Standardisation; **SWL** = Safe Working Load.
Note 3. Legend: (A) Affected by ageing, (G) Affected by gender
Note 4. Units of measure: Hz = Hertz, **mm** = millimetre, **kg** = kilogram, **N** = Newtons

Figure 2 Alt Text: A multi-column summary chart comparing key design and test requirements for walkers and smart-walkers across ISO, EN, and IEC standards. Includes frame loading, stability, braking, durability, ergonomics, and smart-walker use case, as well as user-related factors such as body size, grip strength, and smart-walker classifications.

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Summary of General Features

Handle heights [AD] (n=98)



Handle grips [AD] (n=98)



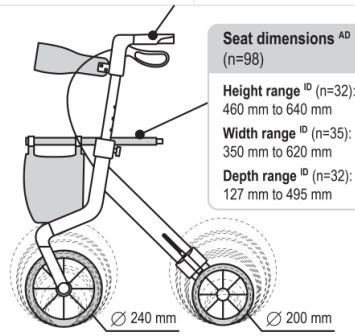
Standard handle (n=76)



Tilted ergonomic handle (n=22)

Height increments [ID]

30 mm (n=3), 25 mm (n=4)
20 mm (n=1), 15 mm (n=1)



Seat dimensions [AD] (n=98)

Height range [ID] (n=32): 460 mm to 640 mm
Width range [ID] (n=35): 350 mm to 620 mm
Depth range [ID] (n=32): 127 mm to 495 mm

Wheel diameters (mm) [ID]

Front (n=104, 18 types)			**Rear** (n=104, 16 types)	
100 (n=1)	112 (n=1)	120 (n=1)	100 (n=1)	112 (n=1)
150 (n=12)	180 (n=3)	190 (n=1)	120 (n=1)	150 (n=12)
200 (n=55)	225 (n=1)	230 (n=2)	180 (n=5)	190 (n=1)
240 (n=6)	245 (n=2)		200 (n=67)	
Off-road wheels:			**Off-road wheels:**	
250 (n=9)	280 (n=3)		225 (n=1)	230 (n=4)
285 (n=1)	300 (n=2)		235 (n=1)	250 (n=1)
305 (n=1)	310 (n=2)		280 (n=1)	300 (n=4)
350 (n=1)			305 (n=1)	310 (n=2)
			350 (n=1)	

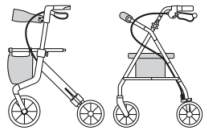
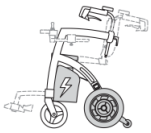
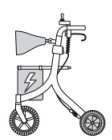
Wheel diameters (mm) [ID]
Mixed front (F), rear (R) diameters (n=22, 8 types)

200 (F), 180 (R), (n=1)	200 (F), 300 (R), (n=2);
240 (F), 200 (R), (n=6)	245 (F), 200 (R), (n=2),
250 (F), 180 (R), (n=1)	250 (F), 200 (R), (n=7)
Off-road wheels:	
280 (F), 230 (R), (n=2)	285 (F), 235 (R), (n=1)

Wheel materials [ID]

Soft (n=28)	**Hard** (n=10)	**Mixed** (n=3)
TPE/TPR (n=20)	PU/PUR (n=9)	PU (front)
EVA (n=1)	PVC (n=1)	TPR (back), (n=2)
Silicone (n=1)		
Pneumatic		PU (front)
Rubber (n=6)		EVA (back), (n=1)

Specifications by Rollator Types

	Brake-grip spans [ID]	Brake mechanisms	Bag capacities [ID]	Device weight [AD]	Device Length and Width
General use (n=72) 	62 mm (n=3) 65 mm (n=1) 70 mm (n=5)	Lever (n=48) Pad (n=28) Disk (n=1) Drum (n=1)	4.0 kg (n=1) 4.5 kg (n=1) 5.0 kg (n=25) 10 kg (n=7) 12 kg (n=2)	**Range:** 4.75 kg to 14 kg **Median:** 7.2 kg	**Length Range:** 495 to 1000 mm **Median:** 662.5 mm **Width Range:** 490 to 770 mm **Median:** 612.5 mm
Indoor use (n=9)  No seat (n=8 of 9)	75 mm (n=1) 77 mm (n=1)	Lever (n=3) Pad (n=3) Foot (n=1)	5.0 kg (n=6)	**Range:** 4.4 kg to 16.5 kg **Median:** 8.1 kg	**Length Range:** 510 to 690 mm **Median:** 631.5 mm **Width Range:** 540 to 680 mm **Median:** 560 mm
Walking trolley (n=2) 	No data	Pad (n=1) Drum (n=1)	12 kg (n=1) 25 kg (n=1)	**Range:** 6.8 kg to 7.7 kg **Median:** 6.9 kg	**Length Range:** 600 to 630 mm **Median:** 615 mm **Width Range:** 570 to 623 mm **Median:** 596.5 mm
Off-road use (n=9) 	62mm (n=3) 70mm (n=5)	Lever (n=4) Pad (n=3) Drum (n=3)	2.0 kg (n=1) 5.0 kg (n=5) 10 kg (n=2) 15 kg (n=1)	**Range:** 6.7 kg to 13.4 kg **Median:** 11 kg	**Length Range:** 730 to 1020 mm **Median:** 815 mm **Width Range:** 570 to 790 mm **Median:** 690 mm
Dual use (n=6) 	No data	Lever (n=5) Drum (n=1)	5 kg (n=1) 10 kg (n=2)	**Range:** 8.1 kg to 11.6 kg **Median:** 9.9 kg	**Length Range:** 630 to 730 mm **Median:** 660 mm **Width Range:** 610 to 690 mm **Median:** 645 mm
Parkinson's walker (n=2) 	No data	Pad (n=1) Drum (n=1)	No data	**Range:** 8.1 kg to 11.6 kg **Median:** 9.9 kg	**Length Range:** 630 to 650 mm **Median:** NA (n=2) **Width Range:** 600 to 670 mm **Median:** NA (n=2)
Powered rollator (n=6) 	No data	Drum (n=1)	No data	**Range:** 12.5 kg to 23 kg **Median:** 17.9 kg	**Length Range:** 630 to 1000 mm **Median:** 710 mm **Width Range:** 585 to 670 mm **Median:** 664 mm

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Figure 3 Caption: Summary of commercial rollator and smart-walker features and specifications

Note 1. Wheel materials: PU/PUR = Polyurethane, **PVC** = Polyvinyl chloride, **TPE/TPR** = Thermoplastic rubbers, **EVA** = Ethylene-vinyl acetate;
Note 2. Legend: [AD] Aggregated data for all rollators. [ID] Incomplete data (Not all products provided information for this criteria);
Note 3. Units of measure: (mm) = millimetre, (**kg)** = kilogram;
Note 4. Instances: Total products reviewed (n=106): Rollators (n=98), Parkinson's smart-walker (n=2), Powered rollators (n=6), Suppliers searched (n=30)
Note 5. Included in **Appendix A** is a full list of rollator and smart walker models classified under general use (n=72), indoor use (n=9), walking trolley (n=2), off-road use (n=9), dual wheelchair or transit use (n=6), Parkinson's disease smart-walker (n-2), and powered rollator (n=6).

Figure 3 Alt Text: Summary of 106 commercial rollators and smart-walkers presenting variations in device dimensions, seat size, brake types, wheel size and material, handle height and style, bag capacity, and device weight. Data is grouped by five rollator and two smart-walker categories, with aggregated statistics by feature and type.

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Appendix A: Commercial rollator and smart-walker features and specifications data

Suppliers (n=30)	Products (n=106)	Walker Type (n) x, (n) x See Notes	Brake (n) Types	Handle (n) Types	Brake Grip Span (mm)	Bag Capacity (kg)
(1) byAcre	(1) Ultra Light (2) Overland (3) Nordic Pioneer (4) Butler HD	(1) R, (1) G (2) R, (1) OR (3) R, (2) G (4) R, (1) ID	(1) Friction contact	(4) Tilted	62 77	5 2.5 5
(2) Rollz	(5) Flex (6) Motion /Performance *(7) Motion (Retrofitted)* *(8) Rhythm (Retrofitted)*	(5) R, (1) WT (6) R, (1) Dual *(1) PR* *(1) SW*	(2) Drum	(7) Tilted		25 10
(3) Airgo	(9) Adventure 8 (10) Comfort Plus (11) Ultra Light 6 (12) Duo Hemi (13) Vista (14) Navigator (15) Fusion	(7) R, (3) G (8) R, (4) G (9) R, (5) G (10) R, (2) Dual (11) R, (6) G (12) R, (3) Dual (13) R, (4) Dual	(3) Lever	Standard		
(4) Alpha	(16) 419 (17) 416/417	(14) R, (7) G (15) R, (8) G	Lever	Standard		
(5) Aspire	(18) Vogue Carbon (19) Vogue Aluminum (20) Adventure (21) Indoor Walker (22) Superlite (23) Deluxe Seat 8” (24) Classic Seat 6” Classic Seat 8” (Same as 6”) (25) Mini Seat Walker (26) XL Walker	(16) R, (9) G (17) R, (10) G (18) R, (11) G (19) R, (2) ID (20) R, (12) G (21) R, (13) G (22) R, (14) G (23) R, (15) G (24) R, (16) G	Friction contact	Standard (8) Tilt Xbar Standard		5
(6) Camino	*(27) Camino*	*(2) PR*		(9) Tilted		
(7) Care Quip	(28) Sidefold HF0470	(25) R, (17) G	Lever	(10) Tilted		
(8) Days	(29) 100 Series (30) Heavy Duty (31) Bariatric HD (32) Supa Mack (33) Transit Rollator (34) Breeze	(26) R, (18) G (27) R, (19) G (28) R, (20) G (29) R, (21) G (30) R, (22) G (31) R, (3) ID	Lever	Standard		5
(9) Drive	(35) Nitro (36) Carbon Fibre (37) X Fold (38) X Fold 10 (39) Diamond (40) Roomba Indoor	(32) R, (23) G (33) R, (24) G (34) R, (25) G (35) R, (26) G (36) R, (5) Dual (37) R, (4) ID	Lever	Standard		5
(10) Eazy	(41) Fold and Go (42) Fold-N-Go HD	(38) R, (27) G (39) R, (28) G	Lever	Standard		5
(11) Elipse	(43) Outdoor Carbon (44) Lite Outdoor (45) Lite / Super Lite (46) Heavy Duty	(40) R, (29) G (41) R, (30) G (42) R, (31) G (43) R, (32) G	Lever (4) Disk Lever	Standard		5
(12) Eurowalker	(47) Euro Walker (48) Prestige	(44) R, (33) G (45) R, (34) G	Lever	Standard (11) Tilted		5
(13) Freedom	(49) Genesis 3	(46) R, (35) G	Lever	(12) Tilted		
(14) Giliani Mobility Tek	*(50) Power Rollator Wheelchair*	*(3) PR*	(5) Electric	Standard		
(15) Hero	(51) Travel Light (52) Outdoor (53) Carbon Lite (54) Carbon Lite HD (55) Fusion	(47) R, (36) G (48) R, (37) G (49) R, (38) G (50) R, (39) G (51) R (6) Dual	Lever	Standard		
(16) Human Care	(56) Leia Ergonomic (57) neXus (58) Carl-Oskar (59) Carl-Oskar Reinf (60) Fellow Classic	(52) R, (40) G (53) R, (41) G (54) R, (2) OR (55) R, (3) OR (56) R, (5) ID	Lever	Standard		5 10
	(61) Alpha 416	(57) R, (42) G	Lever	Standard		

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(17) Max Mobility	Alpha 427 (Same as 416) (62) Alpha 438	 (58) R, (43) G				
(18) Nova	(63) 6" Stnd / HD (64) 8" Stnd / HD (65) 6S / Journey Steel (66) GetGo / Zoom (67) Mighty Mack HD (68) Express (69) Monarch (70) Revo	(59) R, (44) G (60) R, (45) G (61) R, (46) G (62) R, (47) G (63) R, (48) G (64) R, (49) G (65) R, (50) G (66) R, (51) G	Lever	Standard		
(19) Novis	(71) Euro	(67) R, (52) G	Lever	Standard		4.5
(20) PQ	(72) PAB830	(68) R, (53) G	Lever	(13) Tilted		
(21) Redgum	(73) Basic B4204 (74) Comfort Ride	(69) R, (54) G (70) R, (4) OR	Lever	Standard		5 2
(22) Rehearse	(75) Server (76) Server HD (77) Router (78) Space CF (79) Explorer (80) Athlon SL/HD Carbon (81) Laser (82) Pixel	(71) R, (55) G (72) R, (56) G (73) R, (57) G (74) R, (58) G (75) R, (5) OR (76) R, (59) G (77) R, (60) G (78) R, (6) ID	Friction Contact Lever	Standard (14) Tilt Xbar		5 10
(23) Topro	(83) Pegasus (84) Troja Original (85) 5G *(86) Troja Neuro (Retrofitted)* (87) Odessey APO (88) Olympus ATR (89) Hestia	(79) R, (61) G (80) R, (62) G (81) R, (63) G *(2) SW* (82) R, (64) G (83) R, (6) OR (84) R, (7) ID	Friction Contact	(21) Tilted	70 65 70 75	 10 15 5
(24) Trust Care (Let's)	(90) Go Out (91) Go Shopping (92) Fly (93) Move (94) Go /Dream	(85) R, (65) G (86) R, (2) WT (87) R, (66) G (88) R, (67) G (89) R, (8) ID	Friction Other	Standard (22) Tilted		10 12 5
(25) Rebotec	(95) Polo Plus-T	(90) R, (68) G	Lever	Standard		4
(26) Saljol	(96) Carbon Rollator (97) Aluminum (98) Page Indoor HD	(91) R, (69) G (92) R, (70) G (93) R, (9) ID	Friction Other (6) Foot	Standard Standard		5 5
(27) Topgun	*(99) Ezi-Walk*	*(4) PR*	Electric	Standard		
(28) Trionic	(100) Walker 9er (101) Walker 12er (102) Walker 14er (103) Veloped 12er	(94) R, (71) G (95) R, (7) OR (96) R, (8) OR (97) R, (9) OR	Drum	Standard Standard		 5
(29) Thrive	(104) Folding Walker	(98) R, (72) G	Lever	Standard		
(30) Zubu	*(105) Power Rollator* *(106) Power Rollator Wheelchair*	*(5) PR* *(6) PR*	Electric	Standard		

Note 1. General notes: (n) = Instance count; **(n) x, (n) x,** = (Instance count) walker type (see Note 2); Underscored italic walker model = Powered rollators, smart-walkers; Underscored italic wheel diameter = Walkers with mismatched wheel diameters.

Note 2. Instances: Suppliers (n=30), **Total products** (n=106); **Walker types: R** (n=98) = Rollator, **G** (n=72) = General indoor/outdoor rollator, **ID** (n=9) = Indoor rollator, **OR** (n=9) = Offroad rollator; **WT** (n=2) = Walking Trolley; **PR** (n=6) = Powered Rollator, Smart walkers providing guided support, **SW** (n=2) = Smart Walkers (Parkinson's disease, gait monitoring);

Note 3. Units of measure: mm = millimetre, **kg** = kilograms;

Formatting note: Appendix A has been reformatted into two portrait-oriented tables to comply with journal submission requirements. The data continues below.

Appendix A (Continued): Commercial rollator and smart-walker features and specifications data (Continued)

Suppliers (n=30)	Handle Height Range (mm-mm)	Height Increment (mm)	Walker Weight (kg)	Wheel Diameter (n) (mm)	Wheel (n) Materials
(1) byAcre	790-910 890-950	30 30	4.8 6.7	(1)200 /(1)200 (1)250 /(1)250	(1) PUR (1) Rubber

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	835-1023		5.5	(1)225 /(1)225	(2) PUR
	835-975		10	(1)180 /(1)180	(2) Rubber
(2) Rollz	700-800, 800-1000		6.7	(2)200 /(2)200	(3) PUR
	820-940, 870-980		7.7	*(3)200 /(1)300*	(1) PU/EVA (3) Rubber
			11.6		(2) PU/EVA
			11.6	*(4)200 /(2)300*	(4) PUR
(3) Airgo	790-910, 840-990		7.35	(5)200 /(3)200	
	762-914, 840-990		7.8	(6)200 /(4)200	
	760-860		4.75	(1)150 /(1)150	
	810-890, 760-930		8	(7)200 /(5)200	
	900-965		6.8	(8)200 /(6)200	
	760-930		9.5	(9)200 /(7)200	
	760-930		8.5	(10)200 /(8)200	
(4) Alpha	760-980		8.3	(1)230 /(1)230	(1) Silicone
	730-990, 750-990		6	(2)150 /(2)150	
(5) Aspire	695-920, 770-1040		5.2	(11)200 /(9)200	(1) TPE
	785-920		6.1	(12)200 /(10)200	(2) TPE(1) PU/EVA
	900-1035		8.8	*(1)280 /(2)230*	(3) TPE
	860-970		4.4	(3)150 /(3)150	(4) TPE
	740-920		5	(4)150 /(4)150	(5) TPE
	834-989		7	(13)200 /(11)200	(6) TPE
	760-915		6	(5)150 /(5)150	
	817-972		7	(14)200 /(12)200	(7) TPE
	743-905		7	(6)150/(6)150	(8) TPE
	850-982		8	(15)200 /(13)200	(9) TPE
(6) Camino					
(7) Care Quip	800-930		6.8		
(8) Days	775-875, 875-995	25	7.3	(16)200 /(14)200	
	855-975	25	8.8	(7)150 /(7)150	
	830-940, 885-980		13	(17)200 /(15)200	
	750-900		9.4	(18)200 /(16)200	
	800-930		8.8	(1)100/ (1)100	
	835-945		5.5	(2)180 /(2)180	
(9) Drive	710-800		8	*(2)250 /(17)200*	
	840-920		5.5	*(3)250 /(3)180*	
	810-940		7.5	(19)200 /(18)200	
	800-930		9	*(1)240 /(19)200*	
	800-950		10	(20)200 /(20)200	
	814-941		6		
(10) Eazy	834-914		6.3	(8)150 /(8)150	
	797-791		13	(21)200 /(21)200	
(11) Elipse	760-850, 920-1050		6.7	*(2)240 /(22)200*	
	895-1015		10.5	(2)280 /(1)280	
	820-945, 930-1055		8.2	*(3)240 /(23)200*	
	915-1040		9.3	*(4)250 /(24)200*	
(12) Eurowalker	790-920		8	(22)200 /(25)200	
	810-910		9	(5)250 /(26)200	
(13) Freedom	900-1020		9.4	(23)200 /(27)200	
(14) Giliani Mobility Tek	850-950		17.9	(24)200 /(28)200	
(15) Hero	780-900		7.6	(25)200 /(29)200	
	825-950, 935-1060		6.6	*(4)240 /(30)200*	
	845-970, 935-1040		7.3	*(5)240 /(31)200*	
	950-1040		7.8	*(6)240/(32)200*	
	850-970		9.3	(26)200 /(33)200	
(16) Human Care	660-760, 800-970		7.9	(27)200 /(34)200	
	780-860, 830-1010		7.8	(28)200 /(35)200	
	770-950		11.5	(1)300 /(3)300	
	770-950		13.4	(2)300/ (4)300	
	770-950, 820-1000		8.7	(29)200 /(36)200	
(17) Max Mobility	730-980		6	(9)150 /(9)150	
	750-990		6	(3)180 /(4)180	(1) EVA
	840-940		5.7	(30)200 /(37)200	
(18) Nova	800-890, 860-940		8	(10)150 /(10)150	(1) TPR
	810-970		9.5	(31)200 /(38)200	(2) TPR
	755-890		7.6	(11)150 /(11)150	(3) TPR
	825-827, 700-790		7.7	(32)200 /(39)200	(4) TPR
	768-920		14	(33)200/ (40)200	(5) TPR
	845-970		9.5	*(6)250 /(41)200*	(1) PU/TPR
	812-914		11	(34)200 /(42)200	(2) PU/TPR

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	790-915		6	(35)200 /(43)200	
(19) Novis	818-945		6.8	(36)200 /(44)200	
(20) PQ	935-1100	15	9.5	(37)200 /(45)200	(1) PVC
(21) Redgum	840-990		8.7	(38)200 /(46)200	
	900-1025		11	(1)305 /(1)305	
(22) Rehearse	630-790, 740-1020		7.4	(39)200 /(47)200	(5) PUR
	660-860, 740-1020		7.9	(40)200 /(48)200	(10) TPE
	660-860, 740-1020		7.4	(41)200 /(49)200	(6) PUR
	660-860, 800-1020		5.2	(42)200 /(50)200	(7) PUR
	780-900, 910-1120		9.2	*(1)285 /(1)235*	(8) PUR
	630-790, 740-1020		5.2	(43)200 /(51)200	(11) TPE
	770-930		9	(44)200 /(52)200	(12) TPE
	860-970		5.2	(12)150 /(12)150	(13) TPE
(23) Topro	790-976		6.6	(45)200 /(53)200	
	640-790, 780-1000		6.9	(46)200 /(54)200	
	710-860, 815-1000		7.8	(47)200 /(55)200	
	690-860, 780-1000		7.8	(48)200 /(56)200	
	710-860, 815-1000		8.2	(49)200 /(57)200	
	690-865, 760-955		7.9	*(3)280 /(3)230*	
	810-1000		6.9	(1)112 /(1)112	
(24) Trust Care (Let's)	770-950		6.4	*(1)245 /(58)200*	
	760-950		6.9	(50)200 /(59)200	
	785-920		6.8	*(2)245 /(60)200*	(14) TPE
	785-920		6.8	(51)200 /(61)200	
	830-950		6.5		
(25) Rebotec	640-740, 740-920		7.8	(52)200 /(62)200	(15) TPE
(26) Saljol	760-880, 810-1030		5.8	(53)200 /(63)200	
	760-880, 810-1030		6.4	*(7)250 /(64)200*	
	Fixed Height		14.2	(1)120 /(1)120	(16) TPE
(27) Topgun	825-950		13.3	*(54)200 /(5)180*	
(28) Trionic	740-950		9.5	(2(230 /(4)230	(1) PU
	710-860, 760-960		9.2	(1)310 /(1)310	(4) Rubber
	810-1000		9.9	(1)350 /(1)350	(5) Rubber
	750-950, 820-1020		12	(2)310 /(2)310	(6) Rubber
(29) Thrive	800-920, 840-960		6.5	(55)200 /(65)200	
(30) Zubu	830-950		12.5	*(8)250 /(66)200*	
	800-900		18.5	*(9)250 /(67)200*	

Note 1. General notes: (n) = Instance count; **(n) x, (n) x,** = (Instance count) walker type (see Note 2); Underscored italic walker model = Powered rollators, smart-walkers; Underscored italic wheel diameter = Walkers with mismatched wheel diameters.

Note 2. Instances: Suppliers (n=30), **Total products** (n=106); **Walker types: R** (n=98) = Rollator, **G** (n=72) = General indoor/outdoor rollator, **ID** (n=9) = Indoor rollator, **OR** (n=9) = Offroad rollator; **WT** (n=2) = Walking Trolley; **PR** (n=6) = Powered Rollator, Smart walkers providing guided support, **SW** (n=2) = Smart Walkers (Parkinson's disease, gait monitoring);

Note 3. Units of measure: mm = millimetre, **kg** = kilograms;

Formatting note: Appendix A has been reformatted into two portrait-oriented tables to comply with journal submission requirements. The data continues from the previous table.

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Appendix B1: Conformity of commercial rollator and smart-walker instruction manuals to ISO 11199-1:2021 and ISO 11199-2:2021, Section 20.3

Suppliers (n=30)	Total products (n=106) Total documents (n=83) Manuals in the final review (n=70)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	(n) Instances =>		(0)	(65)	(70)	(49)	(61)	(69)	(62)	(40)	(59)	(0)	(2)	(60)	(67)
(1) byAcre	(1) Ultra Light	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(2) Overland/Nordic Pioneer	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(3) Butler HD	R		Y	Y	Y	Y	Y	Y		Y			Y	
(2) Rollz	(4) Flex (WT)	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	(5) Motion Performance (Dual)	R (1) D		Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y
	Motion (Retrofitted, PR) *(6) Rhythm (Retrofitted, SW)*	(2) D R		Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y
(3) Airgo	Adventure 8	N													
	Comfort Plus	N													
	(7) Ultra Light 6	R		Y	Y			Y						Y	Y
	(8) Duo Hemi (Dual)	R		Y	Y			Y						Y	Y
	(9) Vista	R		Y	Y			Y						Y	Y
	(10) Navigator (Dual)	R		Y	Y									Y	Y
	(11) Fusion (Dual)	R		Y	Y			Y						Y	Y
(4) Alpha	(12) 419	R			Y		Y	Y	Y	Y	Y				
	(13) 416/417	R			Y		Y	Y	Y	Y	Y				
(5) Aspire	(14) Vogue Carbon	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(15) Vogue Aluminum	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(16) Adventure	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	Indoor Walker	(3) D													
	Superlite	N													
	(17) Deluxe Seat 8”	R		Y	Y		Y	Y	Y		Y			y	Y
	Classic Seat 6”														
	Classic Seat 8” (Same as 6”)	(4) D													
	Mini Seat Walker	(5) D													
	XL Walker	(6) D													
(6) Camino	*Camino (PR)*	N													
(7) Care Quip	Sidefold HF0470	N													
(8) Days	(18) 100 Series	R		Y	Y		Y	Y	Y	Y	Y				Y
	(19) Heavy Duty	R			Y			Y	Y	Y					Y
	(20) Bariatric HD	R			Y			Y	Y	Y					Y
	Supa Mack	N													
	Transit Rollator	N													
	Breeze	N													
(9) Drive	(21) Nitro	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	(22) Carbon Fibre	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	(23) X Fold	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	X Fold 10	N													
	(24) Diamond (Dual)	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(25) Roomba Indoor	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
(10) Eazy	(26) Fold and Go	R		Y	Y		Y	Y	Y						Y
	(27) Fold-N-Go HD	R		Y	Y		Y	Y	Y						Y
(11) Elipse	(28) Outdoor Carbon	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	(29) Lite Outdoor	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	(30) Lite / Super Lite	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
	(31) Heavy Duty	R		Y	Y	Y	Y	Y	Y	Y	Y			Y	Y
(12) Eurowalker	Euro Walker	N													
	Prestige	N													
(13) Freedom	Genesis 3	N													
(14) Giliani Mobility Tek	*Powered Rollator Wheelchair (PR)*	N													
(15) Hero	(32) Travel Light	R			Y		Y	Y	Y		Y				Y
	(33) Outdoor	R		Y	Y		Y	Y	Y		Y			Y	Y
	(34) Carbon Lite	R		Y	Y		Y	Y	Y		Y			Y	Y
	Carbon Lite HD	N													
	Fusion (Dual)	N													
(16) Human Care	(35) Leia Ergonomic	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(36) neXus	R		Y	Y	Y	Y	Y	Y		Y			Y	Y
	(37) Carl-Oskar	R		Y	Y	Y	Y	Y	Y		Y			Y	Y

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	(38) Carl-Oskar Reinf	R	Y	Y	Y	Y	Y	Y		Y		Y	Y
	(39) Fellow Classic	R	Y	Y	Y	Y	Y	Y		Y		Y	Y
(17) Max Mobility	Alpha 416	N											
	Alpha 427 (Same as 416)	N											
	Alpha 438	N											
(18) Nova	(40) 6" Stnd / HD	R	Y	Y		Y	Y			Y			Y
	8" Stnd / HD	(7) D											
	6S / Journey Steel	N											
	GetGo / Zoom	(8) D											
	Mighty Mack HD	N											
	(41) Express	R	Y	Y			Y					Y	Y
	(42) Monarch	R	Y	Y			Y						Y
	Revo	N											
(19) Novis	(43) Euro	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
(20) PQ	PAB830	N											
(21) Redgum	(44) Basic B4204	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(45) Comfort Ride	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
(22) Rehearse	(46) Server	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(47) Server HD	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(48) Router	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	Space CF	(9) D											
	(49) Explorer	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(50) Athlon SL/HD Carbon	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(51) Laser	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(52) Pixel	R	Y	Y		Y	Y	Y	Y	Y		Y	Y
(23) Topro	(53) Pegasus	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(54) Troja Original	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(55) 5G	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(56) *Troja Neuro (Retrofitted, SW)*	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(57) Odessey APO	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(58) Olympus ATR	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(59) Hestia	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
(24) Trust Care (Let's)	(60) Go Out	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(61) Go Shopping (WT)	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(62) Fly	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(63) Move	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(64) Go /Dream	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
(25) Rebotec	(65) Polo Plus-T	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
(26) Saljol	(66) Carbon Rollator	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	(67) Aluminum	R	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
	Page Indoor HD (Fixed height)	(1) NR											
(27) Topgun	*Ezi-Walk (PR)*	(1) ID											
(28) Trionic	Walker 9er	(10) D											
	Walker 12er	(11) D											
	(68) Walker 14er	R	Y	Y	Y	Y	Y	Y		Y		Y	Y
	(69) Veloped 12er	R	Y	Y	Y	Y	Y	Y		Y		Y	Y
(29) Thrive	(70) Folding Walker	R	Y	Y		Y	Y	Y		Y		Y	Y
(30) Zubu	*Powered Rollator (PR)*	N											
	Powered Rollator Wheelchair (PR)	(2) ID											

Note 1. Instruction manual requirements: The following presents a rephrased overview of requirements in ISO 11199-2: 2021, section 20.3. Refer to the specific standards for more detail;

a) Accessibility of information in formats for visual, reading or cognitive disabilities; **b)** The planned usage, encompassing the target user and environment; **c)** Maximum user mass;

d) The maximum safe weight limit for accessories used to carry loads (containers); **e)** Minimum and maximum height of the rollator; **f)** Maintenance instructions, if applicable;

g) Cleaning: If applicable, provide cleaning method, materials and anti-corrosion measures; **h)** Disinfection: If applicable, provide cleaning method, materials and corrosion details;

i) Overall dimensions (width, length, and height) in mm, mass in kg, preparation for use; if applicable, folding or dismantling;

j) Mass expressed in kg if the device is collapsable or contains removable parts over 10 kg; **k)** Safe of device used in combination with other products and what they are, if applicable;

l) Manufacturer-approved accessories, detachable (spare) parts, and materials; **n)** The location and the type of identification for the unique product identification;

o) Adjustments required before using the rollator and their impact on functionality; **p)** Details on adjustment options and required competency required to carry out the task;

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q) Operation of controls**; r)** Dismantling, re-assembly and information on any removable parts**; s)** Safe grip points for moving and handling components, dismantling, assembly or carrying;

t) Surface temperature warning: changes due to external heat sources; **u)** Safety information: If the device usage or feature poses a hazard (e.g. holes, openings);

v) Safety information: If the device usage poses a hazard (e.g. squeezing; **w)** How to obtain warranty information; **x)** Warning for fall risk (e.g. Incorrect use, do not use the device to transport person)

Note 2. General: a) to x) above corresponds to the data in **Table 4** of the main review paper; **n)** = Instance count;

Note 3. Documents reviewed: Total (n=83); **R** (n=70) = Final rollator documents reviewed, which **includes** (SW, n=2) retrofitted smart-walkers; **D** (n=11) = Duplicate documents; **N** (n=23) = No data; **NR** (n=1) = Not relevant; **ID** (n=2) = Powered rollator documents excluded due to insufficient data; **Y** = Yes, the document conforms the specific ISO 11199-2: 2021 requirements

Formatting note: Appendix B1 has been reformatted into two portrait-oriented tables to comply with journal submission requirements. The data continues below.

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Appendix B1 (Continued): Conformity of commercial rollator and smart-walker instruction manuals to ISO 11199-1:2021 and ISO 11199-2:2021, Section 20.3

Suppliers (n=30)	Total products (n=106) Total documents (n=83) Manuals in the final review (n=70)		(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)	(w)	(x)
	(n) Instances =>		(37)	(69)	(69)	(70)	(62)	(69)	(26)	(62)	(64)	(69)	(66)
(1) byAcre	(1) Ultra Light	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	(2) Overland/Nordic Pioneer	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	(3) Butler HD	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
(2) Rollz	(4) Flex (WT)	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(5) Motion	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	Performance (Dual)	(1) D											
	Motion (Retrofitted, PR)	(2) D											
	(6) Rhythm (Retrofitted, SW)	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
(3) Airgo	Adventure 8	N											
	Comfort Plus	N											
	(7) Ultra Light 6	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(8) Duo Hemi (Dual)	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(9) Vista	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(10) Navigator (Dual)	R		Y	Y	Y	Y	Y			Y	Y	Y
	(11) Fusion (Dual)	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
(4) Alpha	(12) 419	R		Y	Y	Y		Y			Y	Y	Y
	(13) 416/417	R		Y	Y	Y		Y			Y	Y	Y
(5) Aspire	(14) Vogue Carbon	R		Y	Y	Y		Y	Y	Y	Y	Y	Y
	(15) Vogue Aluminum	R		Y	Y	Y		Y	Y	Y	Y	Y	Y
	(16) Adventure	R		Y	Y	Y		Y	Y	Y	Y	Y	Y
	Indoor Walker	(3) D											
	Superlite	N											
	(17) Deluxe Seat 8”	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	Classic Seat 6” Classic Seat 8” (Same as 6”)	(4) D											
	Mini Seat Walker	(5) D											
	XL Walker	(6) D											
(6) Camino	*Camino (PR)*	N											
(7) Care Quip	Sidefold HF0470	N											
(8) Days	(18) 100 Series	R		Y	Y	Y	Y	Y		Y	Y	Y	
	(19) Heavy Duty	R		Y	Y	Y		Y		Y	Y	Y	
	(20) Bariatric HD	R		Y	Y	Y		Y		Y	**Y**	Y	
	Supa Mack	N											
	Transit Rollator	N											
	Breeze	N											
(9) Drive	(21) Nitro	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(22) Carbon Fibre	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(23) X Fold	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	X Fold 10	N											
	(24) Diamond (Dual)	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	(25) Roomba Indoor	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
(10) Eazy	(26) Fold and Go	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	(27) Fold-N-Go HD	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
(11) Elipse	(28) Outdoor Carbon	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(29) Lite Outdoor	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(30) Lite / Super Lite	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(31) Heavy Duty	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
(12) Eurowalker	Euro Walker	N											
	Prestige	N											
(13) Freedom	Genesis 3	N											
(14) Giliani Mobility Tek	*Powered Rollator Wheelchair (PR)*	N											
(15) Hero	(32) Travel Light	R				Y				Y			
	(33) Outdoor	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	(34) Carbon Lite	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	Carbon Lite HD	N											
	Fusion (Dual)	N											
(16) Human Care	(35) Leia Ergonomic	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(36) neXus	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(37) Carl-Oskar	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(38) Carl-Oskar Reinf	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(39) Fellow Classic	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
(17) Max Mobility	Alpha 416	N											
	Alpha 427 (Same as 416)	N											

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	Alpha 438	N											
(18) Nova	(40) 6" Stnd / HD 8" Stnd / HD 6S / Journey Steel GetGo / Zoom Mighty Mack HD	R (7) D N (8) D N	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(41) Express	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(42) Monarch	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	Revo	N											
(19) Novis	(43) Euro	R		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
(20) PQ	PAB830	N											
(21) Redgum	(44) Basic B4204	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
	(45) Comfort Ride	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
(22) Rehearse	(46) Server	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(47) Server HD	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(48) Router	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	Space CF	(9) D											
	(49) Explorer	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(50) Athlon SL/HD Carbon	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(51) Laser	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(52) Pixel	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
(23) Topro	(53) Pegasus	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(54) Troja Original	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(55) 5G	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(56) *Troja Neuro (Retrofitted, SW)*	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(57) Odessey APO	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(58) Olympus ATR	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	(59) Hestia	R	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
(24) Trust Care (Let's)	(60) Go Out	R		Y	Y	Y	Y	Y				Y	Y
	(61) Go Shopping (WT)	R		Y	Y	Y	Y	Y				Y	Y
	(62) Fly	R		Y	Y	Y	Y	Y				Y	Y
	(63) Move	R		Y	Y	Y	Y	Y				Y	Y
	(64) Go /Dream	R		Y	Y	Y	Y	Y				Y	Y
(25) Rebotec	(65) Polo Plus-T	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
(26) Saljol	(66) Carbon Rollator	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(67) Aluminum	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	Page Indoor HD (Fixed height)	(1) NR											
(27) Topgun	*Ezi-Walk (PR)*	(1) ID											
(28) Trionic	Walker 9er	(10) D											
	Walker 12er	(11) D											
	(68) Walker 14er	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
	(69) Veloped 12er	R	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
(29) Thrive	(70) Folding Walker	R		Y	Y	Y	Y	Y		Y	Y	Y	Y
(30) Zubu	*Powered Rollator (PR)*	N											
	Powered Rollator Wheelchair (PR)	(2) ID											

Note 1. Instruction manual requirements: The following presents a rephrased overview of requirements in ISO 11199-2: 2021, section 20.3. Refer to the specific standards for more detail;

a) Accessibility of information in formats for visual, reading or cognitive disabilities; **b)** The planned usage, encompassing the target user and environment; **c)** Maximum user mass;

d) The maximum safe weight limit for accessories used to carry loads (containers); **e)** Minimum and maximum height of the rollator; **f)** Maintenance instructions, if applicable;

g) Cleaning: If applicable, provide cleaning method, materials and anti-corrosion measures; **h)** Disinfection: If applicable, provide cleaning method, materials and corrosion details;

i) Overall dimensions (width, length, and height) in mm, mass in kg, preparation for use; if applicable, folding or dismantling;

j) Mass expressed in kg if the device is collapsable or contains removable parts over 10 kg; **k)** Safe of device used in combination with other products and what they are, if applicable;

l) Manufacturer-approved accessories, detachable (spare) parts, and materials; **n)** The location and the type of identification for the unique product identification;

o) Adjustments required before using the rollator and their impact on functionality; **p)** Details on adjustment options and required competency required to carry out the task;

q) Operation of controls**; r)** Dismantling, re-assembly and information on any removable parts**; s)** Safe grip points for moving and handling components, dismantling, assembly or carrying;

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t) Surface temperature warning: changes due to external heat sources; **u)** Safety information: If the device usage or feature poses a hazard (e.g. holes, openings);

v) Safety information: If the device usage poses a hazard (e.g. squeezing; **w)** How to obtain warranty information; **x)** Warning for fall risk (e.g. Incorrect use, do not use the device to transport person)

Note 2. General: a) to x) above corresponds to the data in **Table 4** of the main review paper; **n)** = Instance count;

Note 3. Documents reviewed: Total (n=83); **R** (n=70) = Final rollator documents reviewed, which **includes** (SW, n=2) retrofitted smart-walkers; **D** (n=11) = Duplicate documents; **N** (n=23) = No data; **NR** (n=1) = Not relevant; **ID** (n=2) = Powered rollator documents excluded due to insufficient data; **Y** = Yes, the document conforms the specific ISO 11199-2: 2021 requirements

Formatting note: Appendix B1 has been reformatted into two portrait-oriented tables to comply with journal submission requirements. The data continues from the previous table.

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Appendix B2: Commercial rollator and smart-walker instruction manuals data, expanded safety and maintenance information and document format

		Safety information					
Suppliers (n=30)	**Manuals Reviewed (n=70)**	**User Hand Height and Positioning**	**Elbow Flexion Angle**	**Slope Limit (Side)**	**Slope Limit (All Around)**	**Slope & Incline Warnings (General)**	**Slope & Incline Warnings (Sitting)**
	(n) Instances >>	(26)	(5)	(5)	(1)	(16)	(28)
(1) byAcre	(1) Ultra Light	Y				Y	
	(2) Overland	Y				Y	
	Nordic Pioneer						
	(3) Butler HD	Y					
(2) Rollz	(4) Flex (WT)	Y					
	(5) Motion	Y		3 Deg.		Y	Y
	Performance (Dual)						
	Motion (Retrofited, PR)						
	(6) Rhythm (Retrofited, SW)	Y		3 Deg.		Y	Y
(3) Airgo	Adventure 8						
	Comfort Plus						
	(7) Ultra Light 6	Y				Y	Y
	(8) Duo Hemi (Dual)	Y				Y	Y
	(9) Vista	Y				Y	Y
	(10) Navigator (Dual)						
	(11) Fusion (Dual)	Y				Y	Y
(4) Alpha	(12) 419						
	(13) 416/417						
(5) Aspire	(14) Vogue Carbon						Y
	(15) Vogue Aluminum						Y
	(16) Adventure						y
	Indoor Walker						
	Superlite						
	(17) Deluxe Seat 8”						
	Classic Seat 6”						
	Classic Seat 8”						
	(Same as 6”)						
	Mini Seat Walker						
	XL Walker						
(6) Camino	*Camino (PR)*						
(7) Care Quip	Sidefold HF0470						
(8) Days	(18) 100 Series	Y	Y				
	(19) Heavy Duty					Y	
	(20) Bariatric HD					Y	
	Supa Mack						
	Transit Rollator						
	Breeze						
(9) Drive	(21) Nitro						
	(22) Carbon Fibre						
	(23) X Fold						
	X Fold 10						
	(24) Diamond (Dual)						
	(25) Roomba Indoor						
(10) Eazy	(26) Fold and Go						
	(27) Fold-N-Go HD						
(11) Elipse	(28) Outdoor Carbon						
	(29) Lite Outdoor						
	(30) Lite / Super Lite						
	(31) Heavy Duty						
(12) Eurowalker	Euro Walker						
	Prestige						
(13) Freedom	Genesis 3						
(14) Giliani Mobility Tek	*Powered Rollator Wheelchair (PR)*						
(15) Hero	(32) Travel Light						
	(33) Outdoor						Y
	(34) Carbon Lite						Y
	Carbon Lite HD						
	Fusion (Dual)						

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(16) Human Care	(35) Leia Ergonomic (36) neXus (37) Carl-Oskar (38) Carl-Oskar Reinf (39) Fellow Classic					Y Y Y	Y Y Y
(17) Max Mobility	Alpha 416 Alpha 427 (Same as 416) Alpha 438						
(18) Nova	(40) 6" Stnd / HD 8" Stnd / HD 6S / Journey Steel GetGo / Zoom Mighty Mack HD (41) Express (42) Monarch Revo	Y Y	Y Y Y				
(19) Novis	(43) Euro	Y			9 Deg	Y	Y
(20) PQ	PAB830						
(21) Redgum	(44) Basic B4204 (45) Comfort Ride	Y	Y			Y Y	Y
(22) Rehearse	(46) Server (47) Server HD (48) Router Space CF (49) Explorer (50) Athlon SL/HD Carbon (51) Laser (52) Pixel						Y Y Y Y Y Y
(23) Topro	(53) Pegasus (54) Troja Original (55) 5G (56) *Troja Neuro (Retrofited, SW)* (57) Odessey (58) Olympus (59) Hestia	Y Y Y Y Y Y Y		5 Deg 5 Deg 5 Deg			Y Y Y Y Y Y
(24) Trust Care (Let's)	(60) Go Out (61) Go Shopping (WT) (62) Fly (63) Move (64) Go /Dream						
(25) Rebotec	(65) Polo Plus-T	Y					
(26) Saljol	(66) Carbon Rollator (67) Aluminum Page Indoor HD (Fixed height)	Y Y					
(27) Topgun	*Ezi-Walk (PR)*						
(28) Trionic	Walker 9er Walker 12er (68) Walker 14er (69) Veloped 12er						
(29) Thrive	(70) Folding Walker	Y					
(30) Zubu	*Powered Rollator (PR)* *Powere Rollator* *Wheelchair (PR)*						

Note. General: See notes in Appendix B-1; **(n)** = Instance count; **Deg** = Degrees; **Y** = Yes, the document conforms to the criteria in the heading

Formatting note: Appendix B2 has been reformatted into two portrait-oriented tables to comply with journal submission requirements. The data continues below.

Appendix B2 (Continued): Commercial rollator and smart-walker instruction manuals data, expanded safety and maintenance information and document format

	Maintenance information	Document format

For Peer Review

Suppliers (n=30)	Manuals Reviewed (n=70)	Wear and tear inspection	Servicing advice	Servicing plan	Booklets (Detailed)	Leaflet (Minimal)	Graphical Instructions
	(n) Instances >>	(57)	(47)	(7)	(66)	(4)	(58)
(1) byAcre	(1) Ultra Light	Y	Y		Y		Y
	(2) Overland	Y	Y		Y		Y
	Nordic Pioneer						
	(3) Butler HD	Y	Y		Y		Y
(2) Rollz	(4) Flex (WT)	Y	Y		Y		Y
	(5) Motion	Y	Y		Y		Y
	Performance (Dual)						
	Motion (Retrofited, PR)						
	(6) Rhythm (Retrofited, SW)	Y	y		Y		Y
(3) Airgo	Adventure 8						
	Comfort Plus						
	(7) Ultra Light 6		Y		Y		Y
	(8) Duo Hemi (Dual)		Y		Y		Y
	(9) Vista		Y		Y		Y
	(10) Navigator (Dual)					Y	
	(11) Fusion (Dual)		Y		Y		Y
(4) Alpha	(12) 419				Y		Y
	(13) 416/417				Y		Y
(5) Aspire	(14) Vogue Carbon	Y	Y	Y	Y		
	(15) Vogue Aluminum	Y	Y	Y	Y		
	(16) Adventure	Y	Y	Y	Y		
	Indoor Walker						
	Superlite						
	(17) Deluxe Seat 8”	Y	Y	Y	Y		Y
	Classic Seat 6”						
	Classic Seat 8”						
	(Same as 6”)						
	Mini Seat Walker						
	XL Walker						
(6) Camino	*Camino (PR)*						
(7) Care Quip	Sidefold HF0470						
(8) Days	(18) 100 Series				Y		Y
	(19) Heavy Duty					Y	
	(20) Bariatric HD					Y	
	Supa Mack						
	Transit Rollator						
	Breeze						
(9) Drive	(21) Nitro				Y		Y
	(22) Carbon Fibre				Y		Y
	(23) X Fold	Y			Y		Y
	X Fold 10						
	(24) Diamond (Dual)	Y			Y		Y
	(25) Roomba Indoor	Y			Y		Y
(10) Eazy	(26) Fold and Go	Y			Y		Y
	(27) Fold-N-Go HD	Y	Y		Y		Y
(11) Elipse	(28) Outdoor Carbon	Y			Y		Y
	(29) Lite Outdoor	Y			Y		Y
	(30) Lite / Super Lite	Y			Y		Y
	(31) Heavy Duty	Y	Y		Y		Y
(12) Eurowalker	Euro Walker						
	Prestige						
(13) Freedom	Genesis 3						
(14) Giliani Mobility Tek	*Powered Rollator Wheelchair (PR)*						
(15) Hero	(32) Travel Light	Y	Y			Y	
	(33) Outdoor	Y			Y		Y
	(34) Carbon Lite	Y			Y		Y
	Carbon Lite HD						
	Fusion (Dual)						
(16) Human Care	(35) Leia Ergonomic	Y	Y		Y		Y
	(36) neXus	Y	Y		Y		Y
	(37) Carl-Oskar	Y	Y		Y		Y
	(38) Carl-Oskar Reinf	Y	Y		Y		Y
	(39) Fellow Classic	Y	Y		Y		Y
	Alpha 416						

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(17) Max Mobility	Alpha 427 (Same as 416) Alpha 438					
(18) Nova	(40) 6" Stnd / HD 8" Stnd / HD 6S / Journey Steel GetGo / Zoom Mighty Mack HD	Y	Y		Y	Y
	(41) Express	Y	Y		Y	Y
	(42) Monarch	Y	Y		Y	Y
	Revo					
(19) Novis	(43) Euro	Y	Y		Y	Y
(20) PQ	PAB830					
(21) Redgum	(44) Basic B4204	Y	Y		Y	Y
	(45) Comfort Ride	Y	Y		Y	Y
(22) Rehearse	(46) Server	Y	Y		Y	
	(47) Server HD	Y	Y		Y	
	(48) Router	Y	Y		Y	
	Space CF					
	(49) Explorer	Y	Y		Y	
	(50) Athlon SL/HD Carbon	Y	Y		Y	
	(51) Laser	Y	Y		Y	Y
	(52) Pixel	Y	Y		Y	Y
(23) Topro	(53) Pegasus	Y	Y		Y	Y
	(54) Troja Original	Y	Y		Y	Y
	(55) 5G	Y	Y		Y	Y
	(56) *Troja Neuro (Retrofited, SW)*	Y	Y		Y	Y
	(57) Odessey	Y	Y		Y	Y
	(58) Olympus	Y	Y		Y	Y
	(59) Hestia	Y	Y		Y	Y
(24) Trust Care (Let's)	(60) Go Out	Y			Y	Y
	(61) Go Shopping (WT)	Y			Y	Y
	(62) Fly	Y			Y	Y
	(63) Move	y			Y	Y
	(64) Go /Dream	Y			Y	Y
(25) Rebotec	(65) Polo Plus-T	Y	Y	Y	Y	Y
(26) Saljol	(66) Carbon Rollator	Y	Y		Y	Y
	(67) Aluminum	Y	Y		Y	Y
	Page Indoor HD (Fixed height)					
(27) Topgun	*Ezi-Walk (PR)*					
(28) Trionic	Walker 9er Walker 12er					
	(68) Walker 14er	Y	Y	Y	Y	Y
	(69) Veloped 12er	Y	Y	Y	Y	Y
(29) Thrive	(70) Folding Walker				Y	Y
(30) Zubu	*Powered Rollator (PR)* *Powere Rollator* *Wheelchair (PR)*					

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